

Asian Banks

After two decades of cost discipline, what's next for Asian banks?

Equities

Asia

Banks

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AI as the next phase of cost efficiency

Asian banks have spent the past two decades relying heavily on cost discipline to offset structural revenue headwinds, with meaningful but incomplete success in our view. As traditional efficiency levers mature, we believe AI now represents the most important opportunity to drive structural cost efficiencies, albeit with likely limited near-term P&L impact as deployments remain incremental. Our key takeaways are threefold: 1) historical cost efficiency gains have cushioned, but not fully offset, revenue pressure; 2) AI's medium-term impact will likely be highly uneven across markets and banks, driven by structural readiness rather than technology availability alone; and 3) despite this, AI optionality appears largely unpriced, with little differentiation at the sector, market or bank level today. Over time, we see scope for AI readiness to emerge as a driver of valuation divergence (please refer to our companion APAC Focus note [here](#)).

Cost efficiency has been the primary offset – but insufficient

Over the past 20 years, banks have used cost efficiency as the dominant lever to defend profitability, with opex-to-assets declining materially across most Asian markets. These gains have been driven by banking consolidation, branch rationalisation, process automation and tighter headcount discipline. Banks have also invested heavily in technology over this period; however, much of this spend has largely focused on maintaining competitiveness and operational stability, with limited evidence of step-change cost savings at the system level. As a result, cost savings have only partially offset structural revenue pressure at an aggregate level.

AI adoption is accelerating, but near-term P&L impact remains limited

AI adoption across Asian banks is accelerating, but current deployments remain narrow and largely siloed, focused on use cases such as customer servicing, compliance monitoring, credit assessments and internal productivity tools. As with past waves of technology investment, near-term financial benefits tend to show up as capacity release, risk reduction or cost avoidance rather than explicit cost cuts. In many cases, productivity gains are reinvested into further digitalisation, data infrastructure or compliance requirements. As a result, evidence of a consistent and material near-term P&L uplift from AI remains mixed, reinforcing our view that the early phase of AI adoption is evolutionary rather than transformative.

Medium term: AI as the scalable efficiency lever

Looking ahead, we see AI as the most meaningful lever to drive structural cost efficiencies over the medium term, particularly given that operations and labour account for roughly 60–70% of banks' total cost bases. We believe successful, end-to-end AI integration has the potential to decouple revenue growth from headcount and physical infrastructure, allowing digitally enabled labour to scale at very low marginal cost. However, the ability to realise these gains will vary materially across markets and institutions, depending on factors such as scale, technology culture, data infrastructure, labour flexibility and competitive urgency. While execution risks are significant, higher AI readiness may help sustain relative cost and valuation advantages even as cyclical tailwinds fade.

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Executive summary

Asian banks are facing structural revenue headwinds driven by demographics-led pressure on net interest income (NII) and persistent challenges in scaling fee income. While operating cost discipline over the past two decades has been meaningful, we think it has not been sufficient to fully offset revenue erosion, raising the key question of whether AI can unlock the next leg of sustainable efficiency gains.

1. Structural revenue pressure is real – and persistent

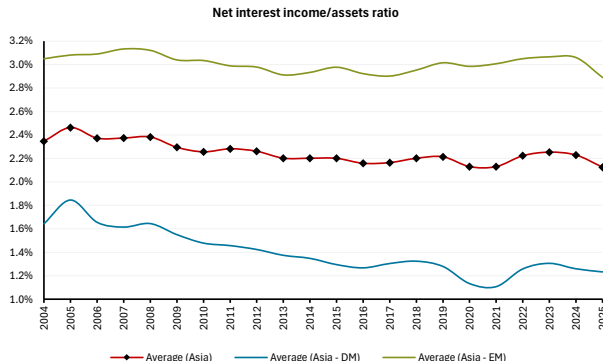
Asian banks are facing increasingly structural revenue headwinds (revenue/assets fell 60bps over the past 20 years), driven primarily by demographic pressures that are weighing on traditional banking income streams.

Ageing populations across the region are slowing deposit accumulation and loan growth, compressing net interest margins and placing net interest income on a long-term downtrend.

Over the past two decades, net interest income as a percentage of assets has declined materially across Asia, while efforts to pivot toward non-interest income have delivered limited relief. Regulatory constraints, structural market features and competitive intensity have made fee income difficult to scale, resulting in a decline in fee income/assets at the regional level.

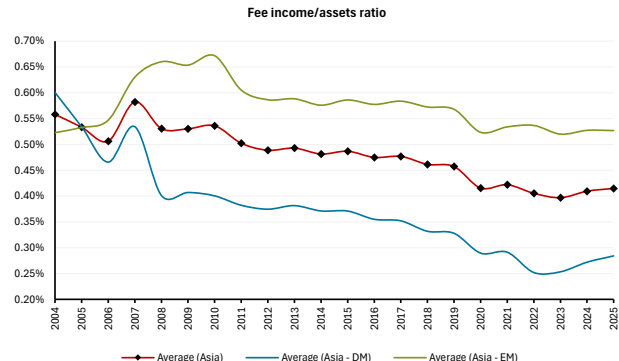
Taken together, total revenue/assets has fallen meaningfully over the past 20 years, reinforcing our view that revenue pressures facing Asian banks are structural rather than cyclical in nature.

Figure 1: Net interest income/assets ratio has declined 45bps on average over the past 20y for Asian banks



Source: Central banks, company data, UBS estimates. Note: JP, AU, KR, SG, HK, TW banks are classified as DM banks above, while IN, CN, ID, PH, TH, MY, VN banks are classified as EM banks.

Figure 2: Fee income/assets has declined by 15bps on average over the past 20y for Asian banks



Source: Central banks, Company data, UBS estimates. Note: JP, AU, KR, SG, HK, TW banks are classified as DM banks above, while IN, CN, ID, PH, TH, MY, VN banks are classified as EM banks.

2. Cost efficiency has helped – but only partially

Against this backdrop, banks have leaned heavily on cost discipline to protect profitability. Operating efficiency has improved steadily over the past two decades, with total operating expenses/assets declining (~40bps over the past 20y) across most Asian markets.

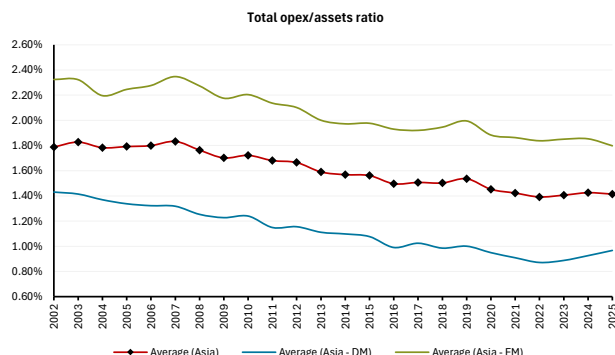
However, these cost savings have only partially offset the revenue decline. At the aggregate level, lower operating expenses have compensated for roughly ⅓ of the fall in revenues, and have not fully insulated bottom-line profitability.

There is also a clear divergence across regions: emerging market banks have, on average, more than offset revenue pressure through cost reductions, while developed

market banks have only been able to offset around ½ of the revenue decline.

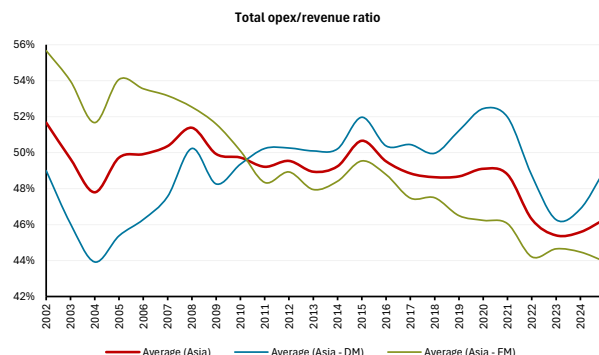
Cost-to-income ratios have converged across Asia, with both developed and emerging market banks now operating at broadly similar levels, contrary to the common perception that emerging market banks are structurally less efficient.

Figure 3: Total opex/assets ratio declined by 40bps on average over the past 20y for Asian banks



Source: Central banks, company data, UBS estimates. Note: JP, AU, KR, SG, HK, TW banks are classified as DM banks above, while IN, CN, ID, PH, TH, MY, VN banks are classified as EM banks.

Figure 4: Cost-to-income ratios for both DM and EM banks have declined over the past 20y



Source: Central banks, company data, UBS estimates. Note: JP, AU, KR, SG, HK, TW banks are classified as DM banks above, while IN, CN, ID, PH, TH, MY, VN banks are classified as EM banks.

3. What drove past cost improvements?

The sources of past efficiency gains suggest many traditional cost levers are now approaching maturity.

Staff costs account for just under half of total operating expenses, and while they have been a key driver of cost reduction in emerging markets, their contribution has been more limited in developed markets.

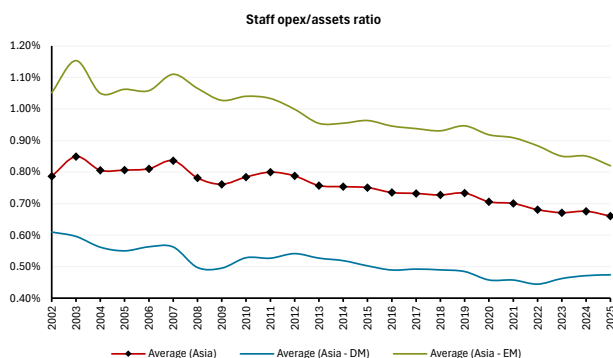
Other operating expenses, including branch, administrative and IT costs, have been the primary source of efficiency gains for developed market banks, reflecting consolidation and branch rationalisation.

While branch networks have been significantly reduced across most Asian markets in recent years, total bank headcount has, interestingly, continued to rise in many countries, likely driven by a shift away from customer-facing roles toward technology, data and back-office functions.

At the same time, technology spending has continued to increase, with Asian banks now spending around 4–5% of revenues on technology on average, and materially more in markets such as Singapore, Australia and Taiwan.

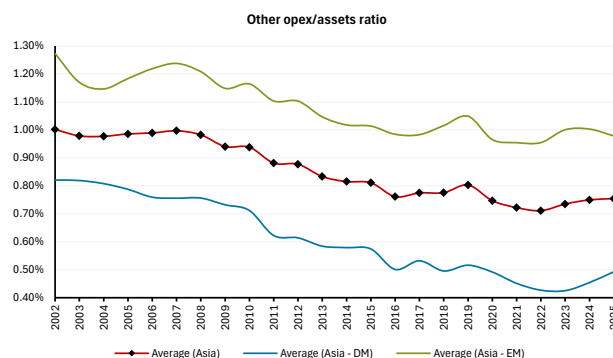
Furthermore, past tech investments have generally not resulted in a big improvement in operational efficiency for the sector. Many banks that increased their tech spend over the past 5 years are yet to see any reduction in their total opex to assets.

Figure 5: Staff opex/assets ratio declined by 15bps over the past 20y for Asian banks



Source: Central banks, company data, UBS estimates. Note: JP, AU, KR, SG, HK, TW banks are classified as DM banks above, while IN, CN, ID, PH, TH, MY, VN banks are classified as EM banks.

Figure 6: Other opex/assets ratio declined by 25bps over the past 20y for Asian banks



Source: Central banks, company data, UBS estimates. Note: JP, AU, KR, SG, HK, TW banks are classified as DM banks above, while IN, CN, ID, PH, TH, MY, VN banks are classified as EM banks.

4. AI adoption: strong momentum, limited near-term P&L impact

AI adoption is accelerating rapidly across Asian banks, but its near-term financial impact remains limited in our view. Banks are increasingly deploying generative and agentic AI across customer engagement, internal productivity and risk management.

However, from a P&L perspective, these initiatives have so far delivered more in the form of capacity release and cost avoidance rather than explicit cost reduction. Productivity gains are often reinvested into further technology modernisation, data infrastructure, cybersecurity and compliance, while improvements in fraud detection and risk analytics primarily enhance earnings resilience rather than driving immediate margin expansion.

Management commentaries across Asian banks consistently highlight operational improvements and strategic importance, but limited short-term impact on reported cost ratios.

5. Medium-term: AI is the largest remaining cost lever

Looking ahead, we believe AI nonetheless represents the most meaningful remaining cost lever for Asian banks over the medium term. Banking operations typically account for 60–70% of the overall cost base, making end-to-end operational AI a potentially powerful driver of efficiency once scaled.

As banks move beyond the initial high-capex phase of building data and AI infrastructure, digital labour can scale at very low marginal cost, allowing revenues to grow without a commensurate increase in headcount or physical infrastructure. Early examples demonstrate the scale of potential operational displacement once AI adoption reaches maturity.

Overall, AI is shifting from a narrative enhancer to a delivery obligation for Asian banks. While near-term financial benefits appear limited, we believe the medium-term opportunity to reshape cost structures is substantial. Execution, rather than intent or messaging, will ultimately determine which banks emerge as the beneficiaries of the next phase of efficiency improvement.

Asian banks are facing structural headwinds on revenue

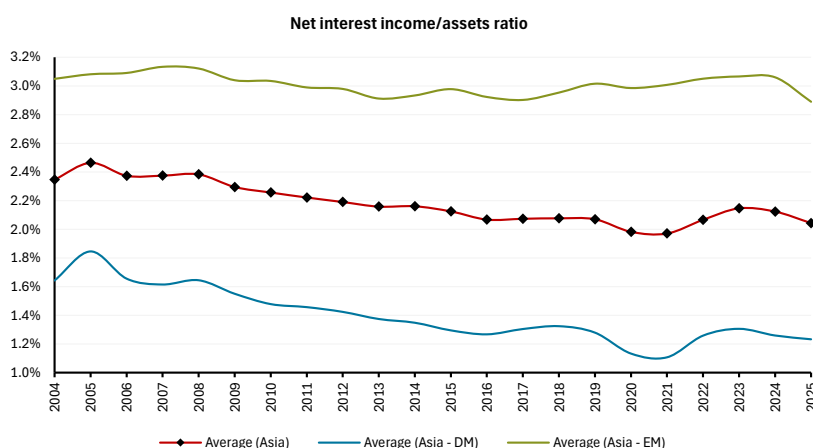
In our Q-Series published last year ([Silver wave: How an ageing population is reshaping Asian finance](#)), we discussed how Asia is undergoing a demographic transformation, with the population anticipated to peak in the next 15 years and by 2050, the population aged 65+ is expected to double. We believe these changes will lead to a rising dependency ratio and have a substantial effect on Asian economies and banking systems.

Net interest income is on a structural downtrend from an ageing demographic

Our analysis identified three significant megatrends stemming from an ageing population. The first is a reduction in consumer savings, followed by a shift in asset allocation from deposits to investments, and lastly, a deceleration in the growth of consumer loans. We believe these megatrends will have a direct effect on Asian banks' profitability and their strategies.

Firstly, as both deposits and lending activities decelerate, with lending experiencing a more pronounced slowdown, **net interest margins (NIM) of these banks are expected to enter a long-term decline, weighing on net interest income (NII).**

Figure 7: Net interest income/assets ratio



Asia banks' net interest income/assets ratio has declined 45bps on average over the past 20y

DM banks saw a 60bps NII/assets ratio decline on average over the past 20y, compared to 20bps for the EM banks

Source: Central banks, company data, UBS estimates. Note: JP, AU, KR, SG, HK, TW banks are classified as DM banks above, while IN, CN, ID, PH, TH, MY, VN banks are classified as EM banks.

As this unfolds, we believe banks will be forced to seek alternative income streams, driving them to shift their focus towards non-interest income (non-II); a trend that has been increasingly playing out over the past few years. In addition, we can expect an increase in banking consolidation. With growing challenges, diminishing profitability, escalating regulatory demands, substantial technology investments, and ongoing pressure on profit margins, we believe the trend towards consolidation in the banking sector is likely to intensify.

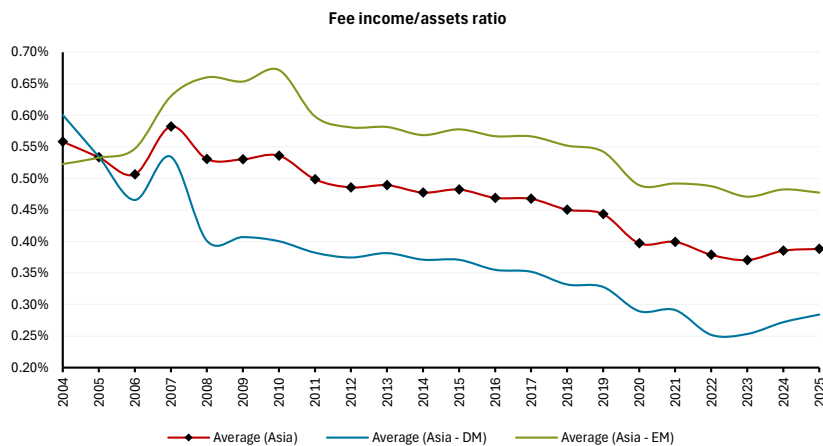
The shift towards fee income to drive revenue growth is also far from straightforward

Our follow-up report last year expanded the discussion around non-II, and specifically on fee income ([The Pivot to Non-interest Income – how challenging is it?](#)).

However, we concluded **the shift towards fee income is far from straightforward.** Many Asian banking markets have faced a combination of regulatory, structural and cyclical challenges over the past couple of decades which negatively impacted their ability to grow their fee income. And, more often than not, we find that focusing on non-II alone is not sufficient to offset the headwinds on net interest income (NII) and

profitability.

Figure 8: Fee income/assets ratio



Source: Central banks, company data, UBS estimates. Note: JP, AU, KR, SG, HK, TW banks are classified as DM banks above, while IN, CN, ID, PH, TH, MY, VN banks are classified as EM banks.

Can lower operating expenses help to offset the headwinds on revenue?

With the structural challenges on NII and headwinds on non-II, can banks turn to improving operating efficiency through the use of technology and AI to lower operating expenses to mitigate the headwinds on revenue?

This is the question we are attempting to answer in this report, where we are expanding the discussion around operating expenses, identifying which markets are doing better/less well, and discuss the challenges/headwinds.

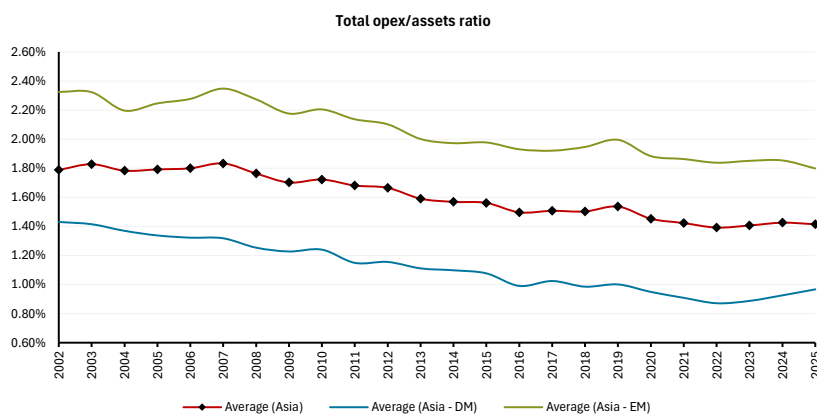
Looking at this firstly on the Asian banks level and the trends over the past 20y, total operating expenses/assets ratio was around 1.41% in 2025 on average (0.97% for DM banks and 1.80% for EM banks), declining by 40bps on average over the past 20y for Asian banks.

In comparison, revenue/assets ratio declined by 60bps on average over this period (-45bps NII/assets ratio and -15bps fee income/assets ratio). **This suggests that lower operating expenses have only helped to offset roughly ⅓ of the decline in revenues over the past 20y, but have not fully offset the impact flowing through to the bottom line.**

With declining net interest income, banks are focusing more on non-interest income/fee income to drive revenue growth. However, that is far from straightforward, as we concluded in our report

Fee income/assets has declined by 15bps on average over the past 20y for Asian banks. DM banks saw a 25bps decline on average over this period, while EM banks declined 5bps

Figure 9: Total operating expenses/assets ratio



Source: Central banks, company data, UBS estimates. Note: JP, AU, KR, SG, HK, TW banks are classified as DM banks above, while IN, CN, ID, PH, TH, MY, VN banks are classified as EM banks.

However, there is an interesting divergence between the DM and EM banks.

Developed market banks' (JP, AU, KR, SG, HK, TW banks) total operating expenses/assets ratio declined by 40bps on average over the past 20y. In comparison, revenue/assets ratio declined by 85bps on average over this period (-60bps NII/assets ratio and -25bps fee income/assets ratio). This suggests that lower operating expenses have only helped to offset roughly 50% of the decline in revenues for DM banks over the past 20y.

However, **emerging market banks'** (IN, CN, ID, PH, TH, MY, VN banks) total operating expenses/assets ratio declined by 45bps on average over the past 20y. In comparison, revenue/assets ratio declined by 25bps on average over this period (-20bps NII/assets ratio and -5bps fee income/assets ratio). This suggests that lower operating expenses have more than offset the decline in revenues over the past 20y, leading to stronger bottom lines for the EM banks.

This divergence suggests emerging market banks have had more scope to extract efficiency gains, while developed market banks may be closer to the frontier of cost optimisation.

What led to the lower opex/assets ratio over the past 20y?

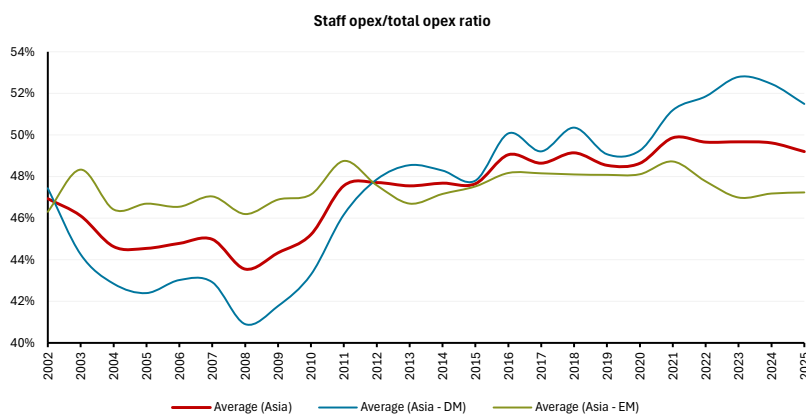
Operating expenses are usually broken down into 2 broad categories for banks – 1) staff costs (e.g salaries, bonuses, employee benefits) and 2) other operating expenses (e.g. administrative expenses, branch and office rent, IT expenses, marketing costs).

Our analysis of the Asian banks suggests staff costs account for slightly less than 50% of total opex (49% of total opex in 2025), though the contribution to total opex could range between 30-65% among markets which we discuss in greater detail in a later section.

Total opex/assets ratio declined by 40bps on average over the past 20y for Asian banks, helping to offset the majority of the decline in revenue/assets ratio, but not sufficient to fully offset the impact to bottom line

Lower operating expenses have more than offset the decline in revenues for EM banks, but unfortunately not so for the DM banks.

Figure 10: Staff opex/total opex ratio



Source: Central banks, company data, UBS estimates. Note: JP, AU, KR, SG, HK, TW banks are classified as DM banks above, while IN, CN, ID, PH, TH, MY, VN banks are classified as EM banks.

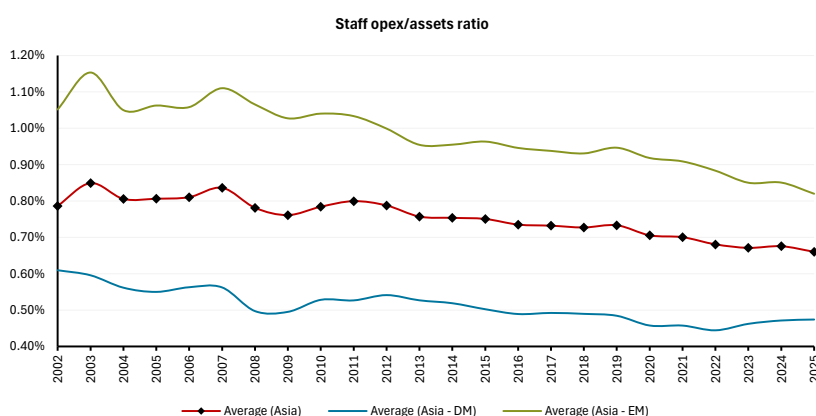
The decline in total opex/assets over the past 20y has been driven by both staff costs and other opex, though we find interesting differences between DM and EM banks.

Firstly on staff costs, staff costs/assets ratio averaged 0.65% in 2025 for Asian banks, declining by 15bps over the past 20y, likely driven to a larger extent by costs cuts given the revenue headwinds, but also, to a smaller extent, likely better operating efficiencies via ongoing technology investments.

Comparing the DM and EM banks, we find their staff opex/assets ratios declined by 10bps and 25bps respectively over the past 20y. Putting this in context of the overall decline in total opex/assets ratio for the 2 categories of banks over the same period (-40bps/-45bps for DM and EM banks respectively), we see the decline in staff costs/assets ratio contributed to 25%/55% of the overall decline in total opex respectively.

In other words, **staff costs contributed to more than half of the decline in total opex for the EM banks over the past 20y, but only about ¼ for the DM banks.**

Figure 11: Staff opex/assets ratio



Source: Central banks, company data, UBS estimates. Note: JP, AU, KR, SG, HK, TW banks are classified as DM banks above, while IN, CN, ID, PH, TH, MY, VN banks are classified as EM banks.

The other opex/assets ratio has also gradually declined over the past 20y. For Asian banks, other opex/assets ratio averaged 0.75% in 2025, declining by 25bps over the past 20y, likely also a reflection of banks trying to offset lower revenues and benefits from technology investments (e.g. branch rationalisation).

Comparing the DM and EM banks, we find their other opex/assets ratios declined by 30bps and 20bps respectively over the past 20y. Putting this in context of the overall decline in total opex/assets ratio over the same period (-40bps for both DM and EM

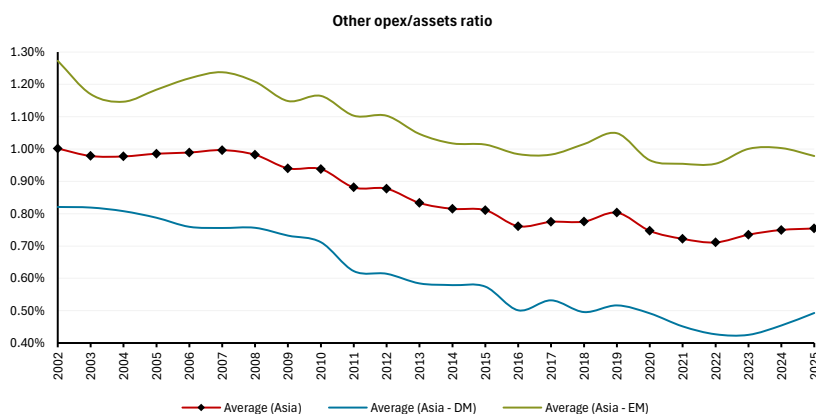
Staff costs accounts for slightly less than 50% of total opex for Asian banks on average

For Asian banks, staff opex/assets ratio declined by 15bps over the past 20y likely driven to a larger extent by costs cuts given the revenue headwinds, but also, to a smaller extent, likely better operating efficiencies via ongoing technology investments

banks), we see the decline in staff costs/assets ratio contributed to 75%/45% of the overall decline in total opex respectively. In other words, **other opex contributed to ¾ of the decline in total opex for the DM banks over the past 20y, but slightly less than half for the EM banks.**

Overall, the composition of cost reduction differs meaningfully across markets, but in both cases, the scope for further gains from traditional levers appears increasingly constrained.

Figure 12: Other opex/assets ratio



For Asian banks, other opex/assets ratio declined by 25bps over the past 20y, with DM banks seeing a relatively larger decline (-30bps) than EM banks (-20bps)

Source: Central banks, company data, UBS estimates. Note: JP, AU, KR, SG, HK, TW banks are classified as DM banks above, while IN, CN, ID, PH, TH, MY, VN banks are classified as EM banks.

Are there significant differences on the cost-to-income ratio?

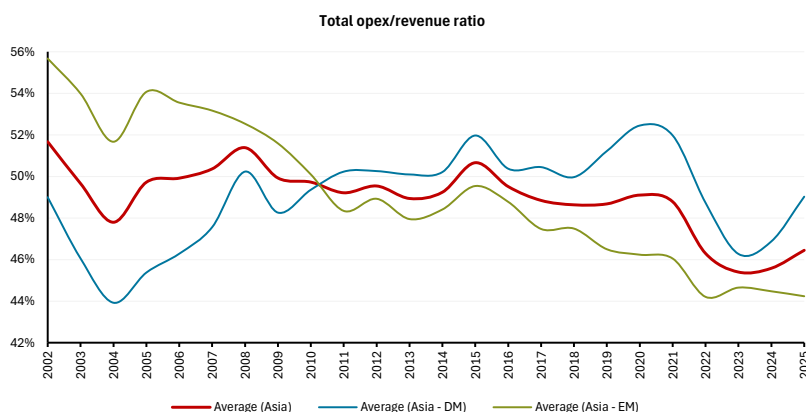
On a cost-to-income (CIR) basis, we find little differences between DM and EM banks, both of which were around the 45-50% range in 2025.

However, EM banks managed to lower their CIR over the past 20y, while CIR increased for the DM banks. On an Asian banks level, CIR was around 46% in 2025, declining 1ppt over the past 20y, driven by a 7ppt decline for the EM banks offsetting a 5ppt increase for the DM banks.

This is interesting, as the common perception has been that EM banks should operate with higher CIR, given their need for extensive branch networks to reach underserved segments, as well as generally lower operating efficiencies due to more limited technological investment. However, our findings challenge this view, with EM banks demonstrating CIRs broadly comparable to those in developed markets.

What explains this apparent divergence from conventional thinking?

Figure 13: Total opex/revenue ratio



Cost-to-income ratios for the both DM and EM banks have declined over the past 20y, and ranged around 45-46% in 2025

Source: Central banks, company data, UBS estimates. Note: JP, AU, KR, SG, HK, TW banks are classified as DM banks above, while IN, CN, ID, PH, TH, MY, VN banks are classified as EM banks.

Why are China, Vietnam, and Hong Kong (domestic) banks able to run at sub-40% cost-income ratio?

We have been asked by several investors in the past why China, Vietnam, and Hong Kong (domestic) banks are able to run at sub-40% cost-income ratio (CIR) when their EM and DM counterparts are running at 44%/49% respectively.

China banks: Chinese banks reported around 38% CIR in 2025. Average CIR from 2014-24 was around 35%. Higher CIR in 2025 was driven by faster growth in costs relative to revenue growth. Nevertheless, CIR was still comparatively lower compared to the EM banks average (38% in China vs 44% in EM). Looking at costs from an ROA perspective, China banks also performed better (ie. lower ratios) on both staff costs/assets and other opex/assets. This could likely be due to China banks' relatively stronger tech and digital capabilities, driving better operating efficiencies and lower costs (measured on both CIR and ROA basis).

Vietnam banks: Vietnamese banks recorded around 30% CIR in 2025, steadily declined from mid-40% in 2015. We believe this is a reflection of stronger revenue growth (high credit growth and fee growth, offsetting NIM pressure) relative to costs growth. Not only are Vietnamese banks performing better on CIR (compared to EM), they also outperformed (ie. lower ratios) on both staff costs/assets and other opex/assets, albeit less so than Chinese banks. Total opex/assets ratios were 1.3% in Vietnam, compared to 0.7% in China and 1.8% for EM banks. This suggests the Vietnam banks' relatively lower CIR of 30% was attributed more to stronger revenues (compared to EM), but likely relatively less due to tech and digital capabilities as with the Chinese banks.

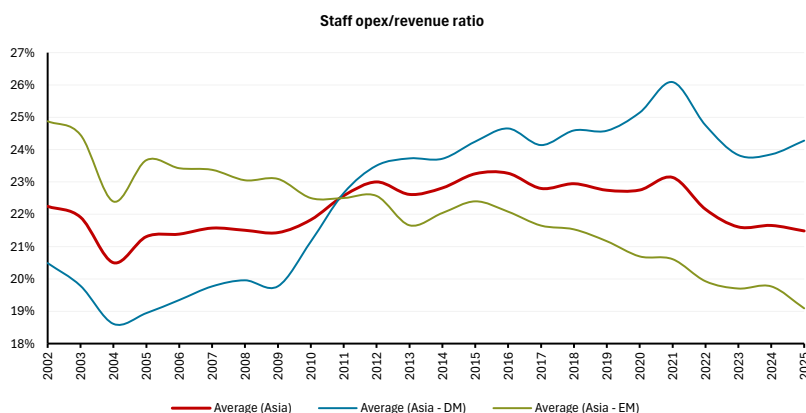
Hong Kong banks: The key driver of the lower CIR is attributed to BOCHK, which reported a 24% CIR 2025. The other HK (domestic) bank, BEA, reported a CIR of 47% in 2025 which is close to the DM average. BOCHK has also impressively achieved positive JAWS with over the past few years with CIR averaging 29% over 2014-24. We attribute the strong operating efficiency to the bank's scale in Hong Kong (16-17% market share), and costs optimisation efforts by centralizing back-office operations in lower-cost regional hubs such as Nanning and deploying AI and robotic process automation to maximise back-office efficiency.

Diving deeper into the opex breakdown, DM banks' staff costs/revenue ratio was roughly 24% on average in 2025, while this was slightly lower for EM banks at around 19%.

This is an interesting observation as the general view has been that DM banks should have lower staff costs/revenue ratio as they have invested more in technology which should yield better operating efficiencies and lower staff count (through increased automation and lesser need for branches). However, this finding appears to be at odds with the common perception whereby **DM banks are actually running at higher staff costs/revenue ratio than the EM banks.**

We think this could be due to a few possible reasons: 1) higher wages in DM markets relative to EM markets which offsets the lower staff count in general, 2) costs savings on lower administrative staff count is replaced by higher number of tech employees which generally earn higher wages, and 3) technological investments takes time to yield significant costs savings given the legacy banking systems which have to be replaced.

Figure 14: Staff costs/revenue ratio



Staff counts accounted for 22% of revenue for DM banks on average in 2025, slightly higher than 20% for the EM banks

Source: Central banks, company data, UBS estimates. Note: JP, AU, KR, SG, HK, TW banks are classified as DM banks above, while IN, CN, ID, PH, TH, MY, VN banks are classified as EM banks.

Banking consolidation has helped in lowering cost pressures

With increasing revenue pressures facing the banking sector, and the need for continued and higher costs investments (higher tech and regulatory costs), **we have seen banking consolidation in several markets over the past 20y.**

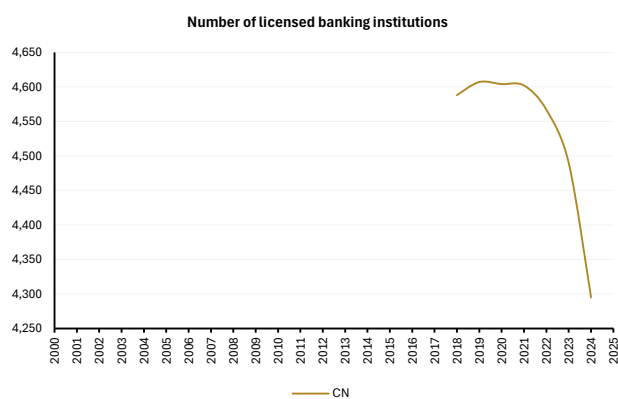
China has the largest number of licensed banking institutions among the Asian banking markets with close to 4,300 banks as of 2024 (-6% since 2018).

Indonesia has the second highest number of banks with close to 1,500 banks as of 2024, despite the total count down around 25% over the past 20y. We also note how the total bank count in the Philippines has halved over the past 25y to slightly more than 450 banks in 2025.

Other markets with a high number of banking institutions are India (208 scheduled banks), Hong Kong (149 banks), Singapore (133 banks), and Japan (107 banks). *Note: Definition of banking institutions may vary across markets – definitions can be found in the grey text box below.*

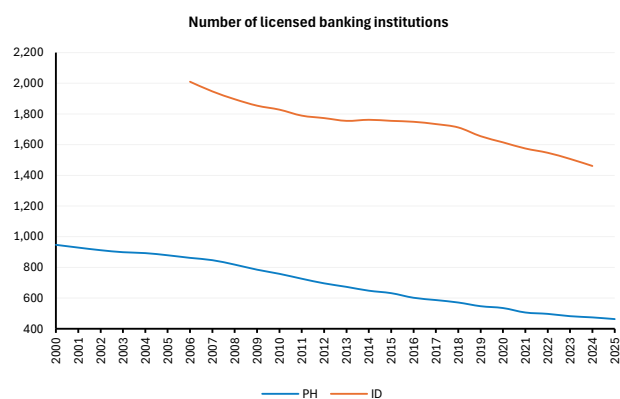
Nevertheless, the key insight in this section is that in most markets the number of licensed banking institutions was generally stable to declining over the past 5 years. Markets that saw a relatively larger decline in banks were the Philippines (-13% since 2020), China, Hong Kong, Australia, and Thailand (all -7% over the past 5y).

Figure 15: Number of licensed banks – CN



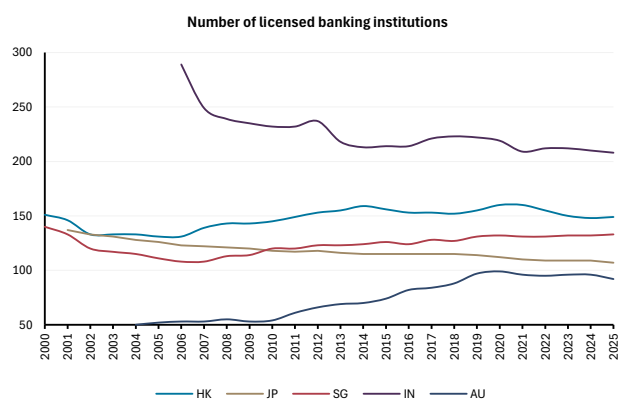
Source: National Financial Regulatory Administration

Figure 16: Number of licensed banks – PH / ID



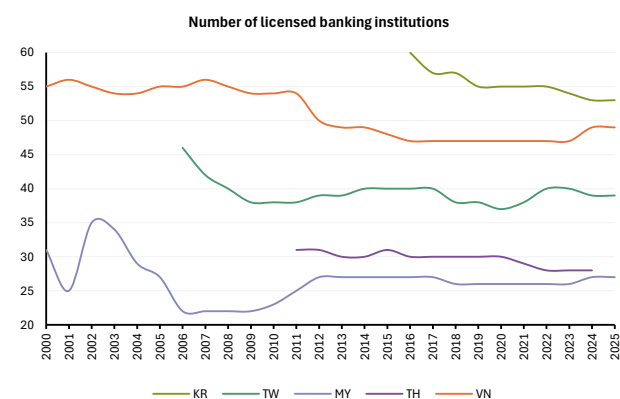
Source: BSP, Indonesia Financial Services Authority

Figure 17: Number of licensed banks – HK / JP / SG / IN / AU



Source: Central banks, Australian Prudential Regulation Authority

Figure 18: Number of licensed banks – KR / TW / MY / TH / VN



Source: Central banks

Definition of banking institutions

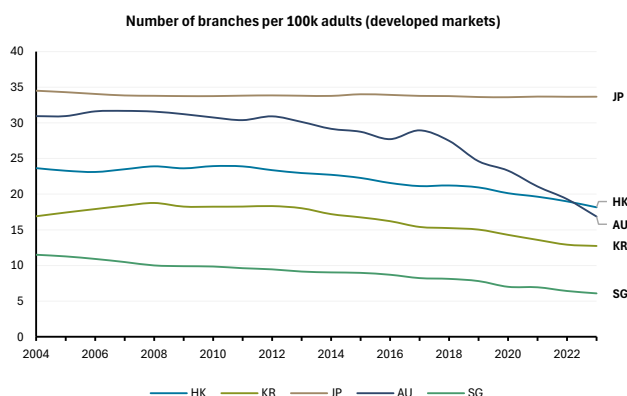
- **China:** Includes City Commercial banks, Privately-owned banks, Rural Commercial banks, Rural Cooperative banks, Rural Credit Cooperatives, Village and Township banks, Direct banks, and Foreign banks.
- **Philippines:** Includes Universal and Commercial banks, Thrift banks, Rural and Cooperative banks, and Digital Banks.
- **Hong Kong:** Includes banks incorporated in Hong Kong and outside of Hong Kong. Licensed banks are primarily allowed to 1) Operate current and savings accounts; 2) Accept deposits of any size and maturity from the public.
- **Japan:** Includes City banks, Trust banks, and Regional banks.
- **Singapore:** Includes both full banks and wholesale banks (restricted from offering retail banking services).
- **Indonesia:** Includes both commercial banks and rural banks. Rural banks accounts for slightly more than 90% of total banks.
- **India:** Includes scheduled commercial banks and scheduled co-operative banks. Scheduled banks refer to banks that are eligible for RBI facilities such as liquidity support and clearing access.
- **Australia:** Includes all authorised deposit-taking institutions (such as banks, building societies and credit unions).
- **Korea:** Includes domestic banks and foreign banks.
- **Taiwan:** Includes only domestic banks.
- **Malaysia:** Only includes commercial banks (both domestic and foreign). Excludes Islamic banks and investment banks.
- **Thailand:** Only includes commercial banks (both domestic and foreign).
- **Vietnam:** Includes policy banks, State-owned Commercial Banks, Joint-stock Commercial Banks, Joint-Venture Banks, and 100% Foreign-Owned Banks.

Branch rationalisation initiatives are also underway

Based on data from the World Bank (latest data as of 2023), **most markets in Asia have seen a decline in branch penetration (measured by the number of branches per 100k adults) from 2018-23, apart from India (+1%) and Japan (flat),** which likely contributed to savings on total opex. Markets that saw the largest decline in branch penetration from 2018-23 were Australia (-39%), Indonesia (-27%), Thailand (-26%), and Singapore (-25%).

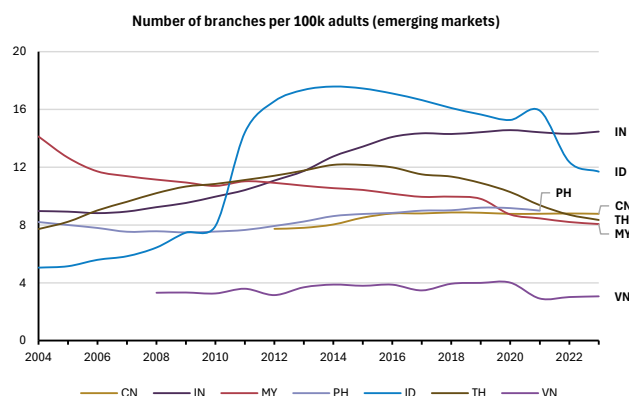
On an absolute basis, branch penetration was the highest in Japan (34 branches per 100k adults) in 2023, followed closely by Hong Kong (18 branches) and Australia (17 branches). On the other end, Vietnam has the lowest branch penetration (3 branches per 100k adults), followed by Singapore (6 branches), and Thailand (8 branches).

Figure 19: Number of branches per 100k adults – HK / KR / JP / AU / SG



Source: World Bank

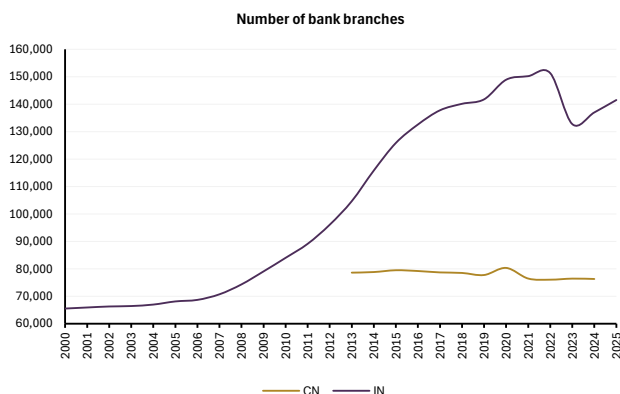
Figure 20: Number of branches per 100k adults – CN / IN / MY / PH / ID / TH / VN



Source: World Bank

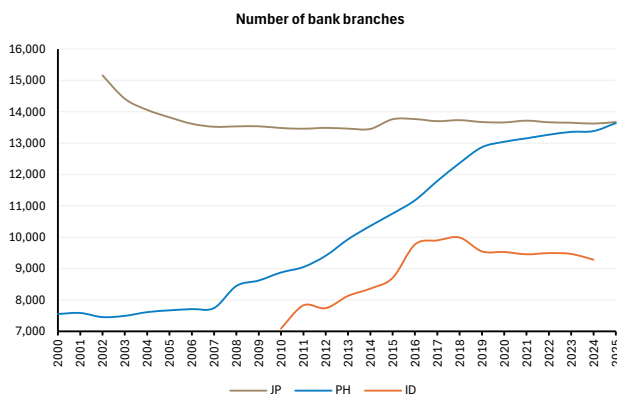
In terms of absolute branch count, all Asian markets, apart from the Philippines and Vietnam, saw a decline in the total number of branches over the past 5y. Markets that saw the largest decline in the absolute number of branches over this period were Australia (-33%), Thailand (-24%) and Korea (-14%). Total number of branches increased 5% over the past 5y in the Philippines, and 7% in Vietnam.

Figure 21: Number of bank branches – IN / CN



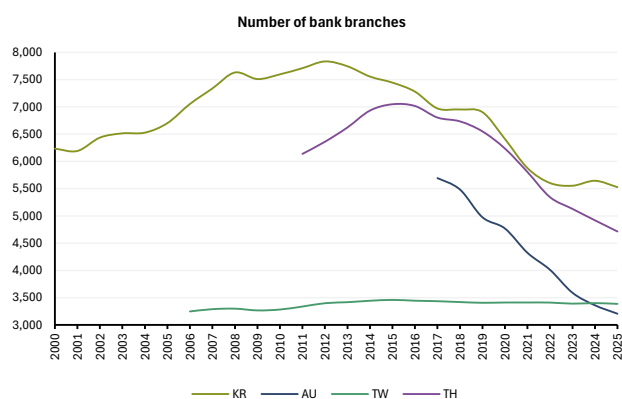
Source: Central banks, company data. Note: Data for China is based on total branch count for ICBC, CCB, ABC, BOC, and BOCOM.

Figure 22: Number of bank branches – JP / PH / ID



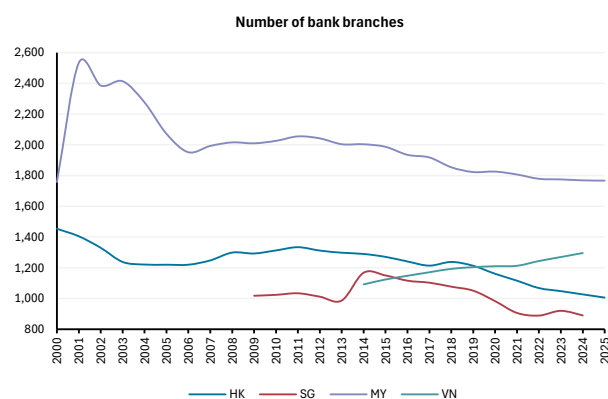
Source: Central banks, Japanese Bankers Association, Indonesia Financial Services Authority.

Figure 23: Number of bank branches – KR / AU / TW / TH



Source: Central banks, Australian Prudential Regulation Authority.

Figure 24: Number of bank branches – HK / MY / SG / VN



Source: Central banks, company data. Note: Data for Singapore is based on total branch count for OCBC and UOB. Data for Vietnam is based on total branch count for TCB, MBB, VPB, and ACB.

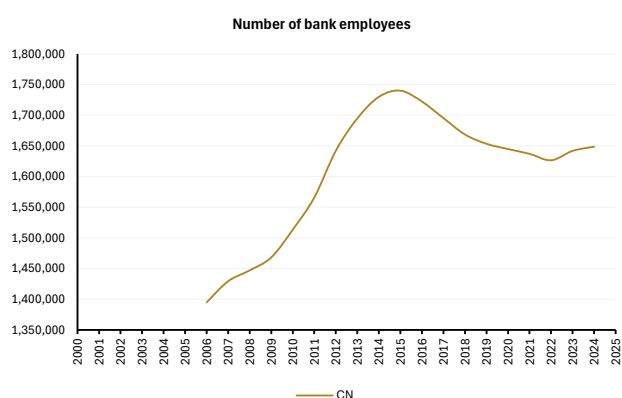
Even as total employee headcount has generally risen...

While the total number of branches in most Asian markets have declined over the past 5y, the total number of bank employees have increased for most markets.

This could have been driven by a shift in employee mix from customer-facing branch tellers in the past to increasingly more back-end tech support staff over the years. Furthermore, the increase in overall headcount could have been driven by the need for more technological investments and thus higher tech headcount among banks over the past few years, given increasing competition from fintechs and increasing demand from end consumers for better digital offerings.

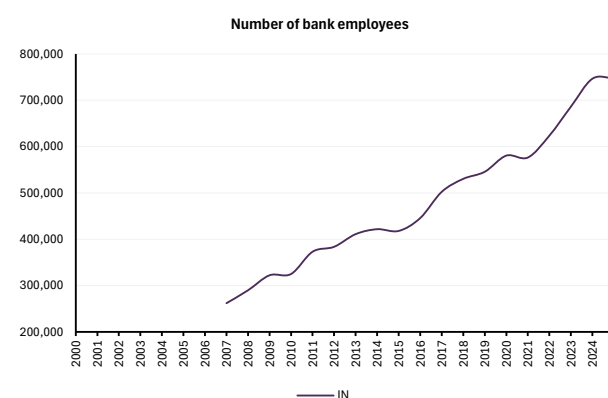
Markets that saw the largest increase in bank employees over the past 5y were India (+37%), Singapore (+25%), and Taiwan (+18%). On the other hand, markets that saw a decline in bank employees were Thailand (-16%), Hong Kong (-5%), and Korea (-5%).

Figure 25: Number of bank employees – CN



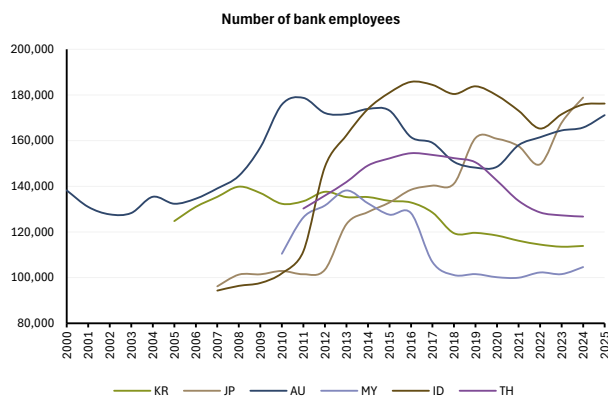
Source: Company data. Note: Data for China is based on total employee count for ICBC, CCB, ABC, BOC, and BCOM.

Figure 26: Number of bank employees – IN



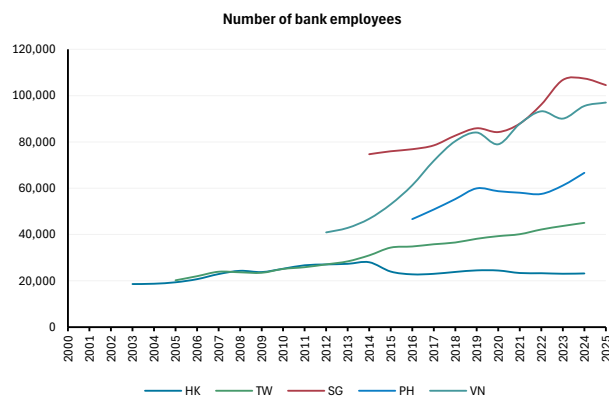
Source: Company data. Note: Data for India is based on total employee count for HDFCB, ICICI, SBI, Axis, and Kotak.

Figure 27: Number of bank employees – KR / JP / AU / MY / ID / TH



Source: Central banks, company data. Note: Data for Japan (MUFG, SMBC), Australia (CBA, NAB, ANZ, Westpac), Indonesia (BBCA, BBRI, BMRI, BNNI).

Figure 28: Number of bank employees – HK / TW / SG / PH / VN



Source: Company data. Note: Data for Hong Kong (BOCHK, BEA), Taiwan (CTBC, Mega, E.Sun), Singapore (DBS, OCBC, UOB), Philippines (BDO, BPI), Vietnam (VCB, TCB, VPB, MBB, ACB).

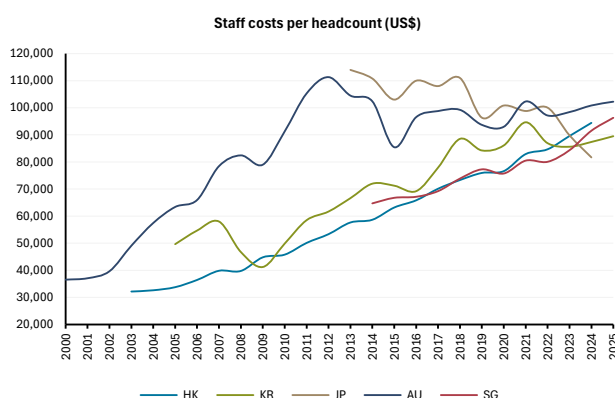
...growth in staff costs per employee has lagged nominal GDP growth

Consistent with the discussion above, even as total bank employee headcount has generally risen across most markets, staff costs per employee have also increased over the past five years but at a pace below nominal GDP growth ([Figure 33](#)).

This suggests improving cost efficiencies, as growth in personnel expenses has been relatively contained despite expanding workforces – likely reflecting a shift toward lower cost back-end and technology-focused roles enabled by higher technology investment, which we discuss further in the following section.

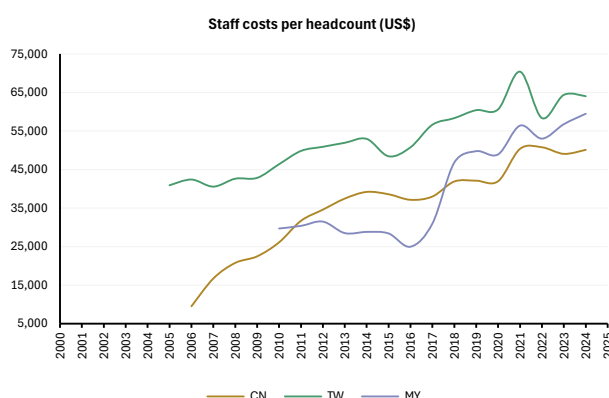
Over the same period, markets that saw the largest increases in staff costs per employee (in US\$ terms) were Hong Kong, Malaysia, Singapore, and China (+5%). Conversely, staff costs per employee rose the least in Japan (-5%), Korea, Thailand, and Taiwan (+1%).

Figure 29: Staff costs per employee – HK / KR / JP / AU / SG



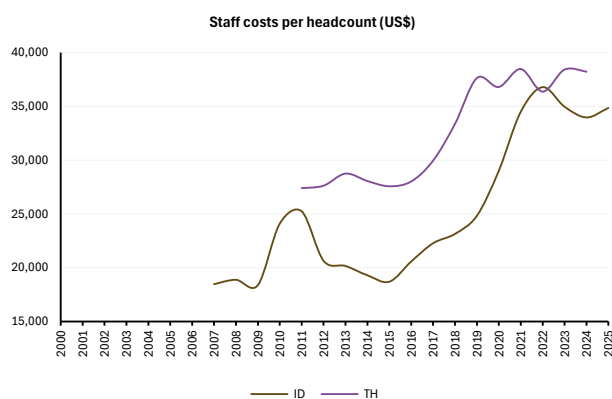
Source: Company data. Note: Data for Hong Kong (BOCHK, BEA), Japan (MUFG, SMBC), Australia (CBA, NAB, ANZ, Westpac), Singapore (DBS, OCBC, UOB).

Figure 30: Staff costs per employee – CN / TW / MY



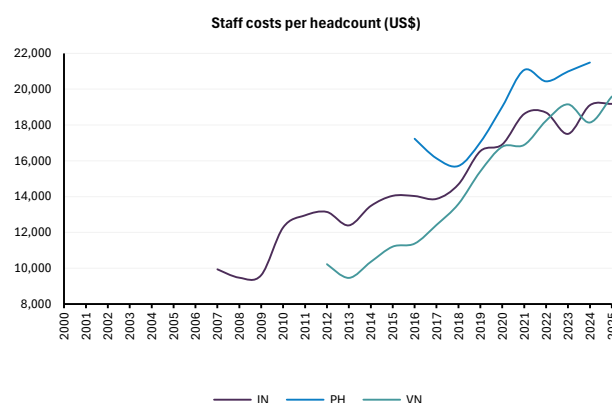
Source: Company data. Note: Data for China (ICBC, CCB, ABC, BOC, and BOCOM), Taiwan (CTBC, Mega, E.Sun).

Figure 31: Staff costs per employee – ID / TH



Source: Company data. Note: Data for Indonesia (BBCA, BBRI, BMRI, BBNi).

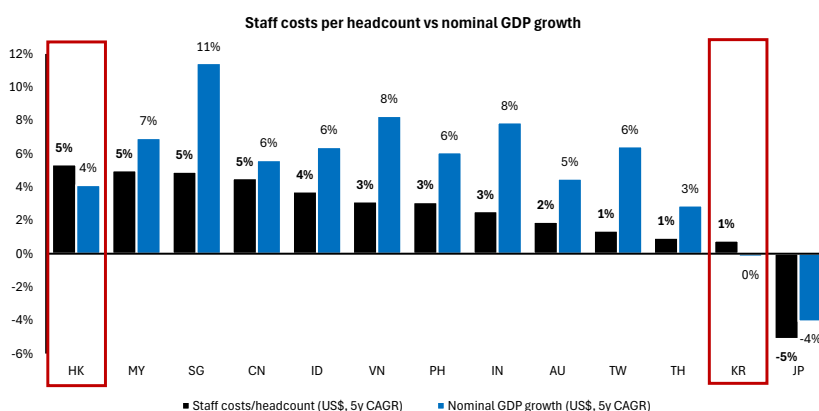
Figure 32: Staff costs per employee – IN / PH / VN



Source: Company data. Note: Data for India (HDFCB, ICICI, SBI, Axis, and Kotak), Philippines (BDO, BPI), Vietnam (VCB, TCB, VPB, MBB, ACB).

While staff costs per employee growth generally lagged nominal GDP growth across the region, we note Hong Kong and Korea were the only 2 markets where staff costs per employee growth outpaced nominal GDP growth. This may point to more limited staff cost gains from technology investments in Hong Kong and Korea, where opportunities to lower average personnel costs through workforce shuffling appear more constrained.

Figure 33: Growth in staff costs per employee relative to nominal GDP growth



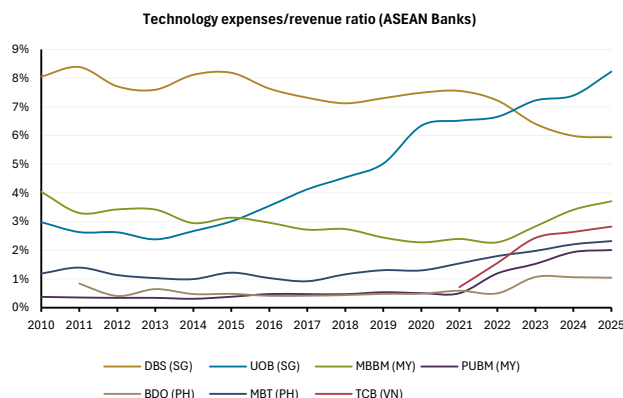
Source: Central banks, company data

How much are banks spending on technology?

Based on company disclosures where available, we calculate the big banks in Asia are **spending around 4-5% of their revenue on technology expenses on average**. This ratio has increased by 1-2ppt on average over the past 10y.

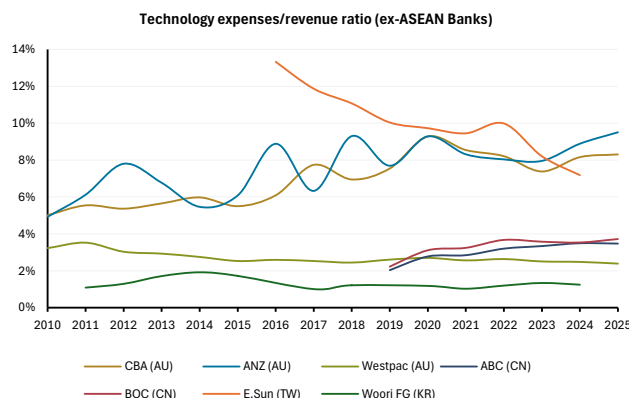
Banks in Singapore, Australia, and Taiwan are spending around 6-10% of their revenues on technology expenses, while in EM markets such as China, Malaysia, Philippines, and Vietnam, technology spend is relatively lower at 1-4% of revenue.

Figure 34: Tech expenses/revenue ratio (ASEAN Banks)



Source: Company data

Figure 35: Tech expenses/revenue ratio (ex-ASEAN Banks)

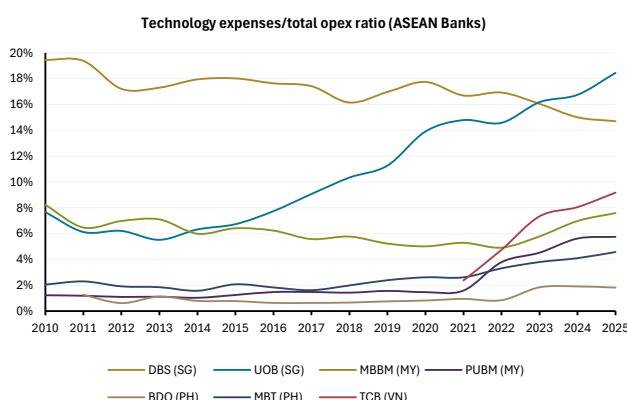


Source: Company data

We calculate **technology expenses accounted for around 9-10% of total opex for the big banks in Asia in 2025**. Markets that stand out are Singapore, Australia, and Taiwan where technology expenses account for 15-18% of total opex. China and Vietnam banks are not far behind, with technology expenses accounting for c10% of total opex.

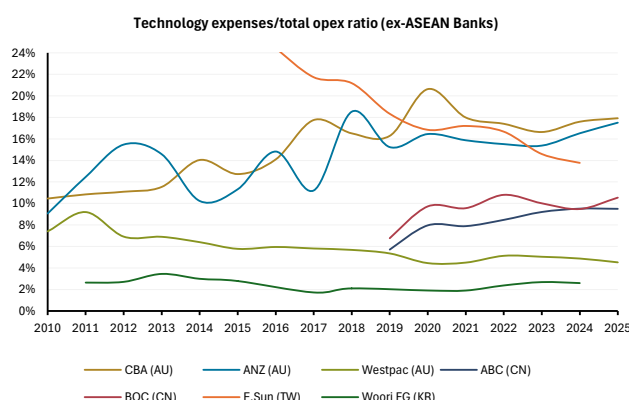
Despite rising technology investment, the link to measurable cost efficiency remains weak at the aggregate level, highlighting the limitations of traditional digitalisation efforts, in our view.

Figure 36: Tech expenses/total opex ratio (ASEAN Banks)



Source: Company data

Figure 37: Tech expenses/total opex ratio (ex-ASEAN Banks)

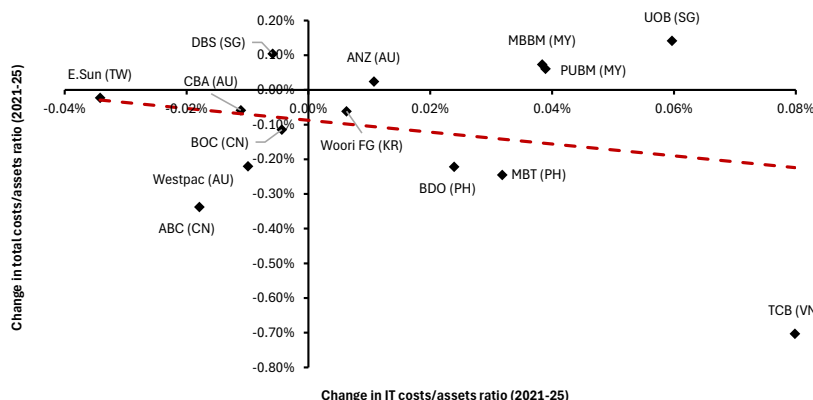


Source: Company data

Are the technology investments leading to costs savings?

Our findings suggest this relationship looks fairly mixed for Asian banks overall. The scatter plot in [Figure 38](#) indicates a very weak negative correlation between technology spend (as a percentage of assets) and total operating expenses (as a percentage of assets). In other words, banks with higher IT costs relative to assets tend to exhibit lower overall cost intensity.

Figure 38: Are higher tech investments leading to costs savings?



Source: Company data

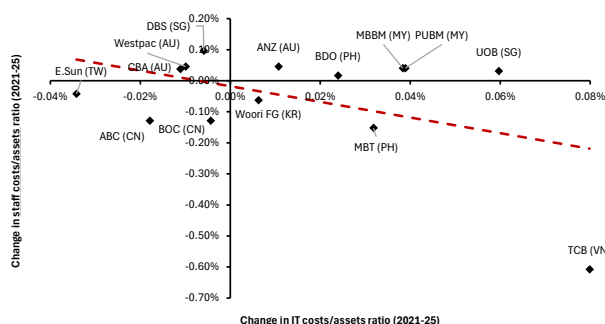
This relationship appears to be more driven by changes in staff cost structure (Figure 39). Technology investments reduce reliance on higher-cost front-end roles by automating customer-facing and manual processes, even as overall headcount continues to grow (see the section above). The resulting shift toward more back-end and operational roles, which are typically lower-cost, helps contain personnel expenses and contributes to lower overall cost intensity.

This relationship is more pronounced in emerging markets such as Vietnam and the Philippines. This is intuitive, as these markets generally started with lower operating efficiency, leaving greater scope to capture meaningful cost savings through higher technology investments.

We believe that continued technology investment should further enhance operating efficiency, particularly across emerging markets, over the medium term. To date, most tech spending has been directed toward digitalising legacy systems, enhancing digital capabilities (e.g. retail and consumer banking interfaces), and strengthening cybersecurity. Looking ahead, we see artificial intelligence as the next potential driver of operating efficiency gains.

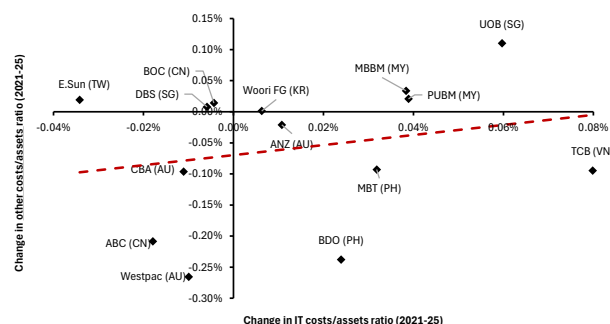
Over time, we believe AI is likely to emerge as a key differentiator among banks: those that invest early and meaningfully in AI, and, importantly, foster an organisational culture that supports experimentation and rapid implementation, are likely to be best positioned to translate technological capability into sustainable efficiency gains and competitive advantage. We discuss this in more detail in the section below – [Will AI drive better costs efficiencies for the Asian banks?](#)

Figure 39: Higher tech investments has generally led to lower staff costs...



Source: Company data

Figure 40: ...but an increase in other costs (which includes tech costs)



Source: Company data

Alternate measures of operating efficiency

Another measure of operating efficiency, aside from the more commonly used cost-to-income ratio (CIR), is the level of loans and deposits per branch, which serves as a proxy for branch-level productivity.

Loans per branch ratio

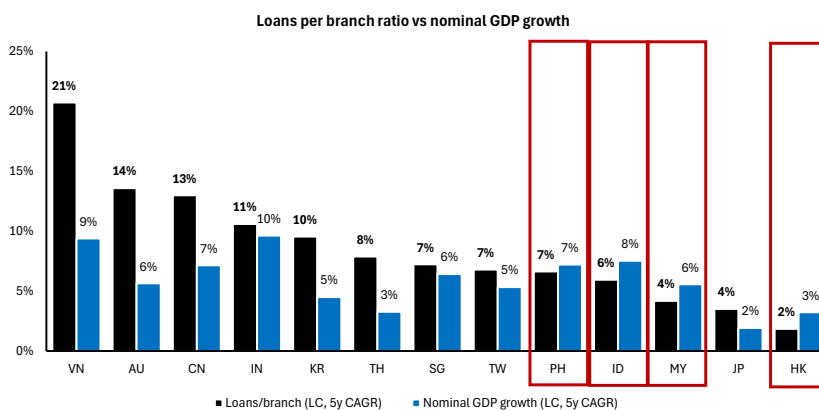
Loans per branch, measured on a LC basis, have increased across all Asian banking markets over the past five years. The largest increases were observed in Vietnam (+21% CAGR), Australia (+14%), China (+13%), and India (+11%).

While loans per branch have risen across the region, **not all markets delivered improvements relative to nominal GDP growth**. This distinction is important, as loan growth is fundamentally driven by nominal GDP; therefore, genuine operating efficiency gains should manifest as faster growth in loans per branch relative to nominal GDP.

Our analysis shows that only around 70% of Asian banking markets recorded loans-per-branch growth that outpaced nominal GDP growth over the past five years – namely Vietnam, Australia, China, India, Korea, Thailand, Singapore, Taiwan, and Japan.

This suggests that banks in these markets were more successful in translating operational improvements (including digitalisation and productivity gains) into higher branch level efficiency on this metric.

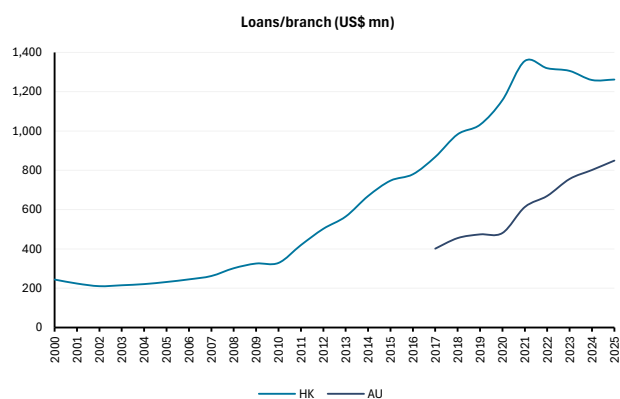
Figure 41: Growth in loans per branch relative to nominal GDP growth



Source: Central banks, Company data.

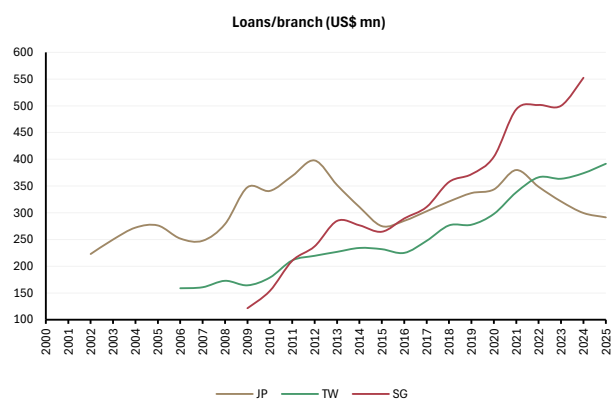
Below charts shows the absolute loans per branch trend for the various Asian markets in US\$ terms. On an absolute basis, it is not surprising to see that the DM banks have higher loans per branch ratio compared to the EM banks given their relatively higher GDP per capita.

Figure 42: Loans per branch (US\$m) – HK / AU



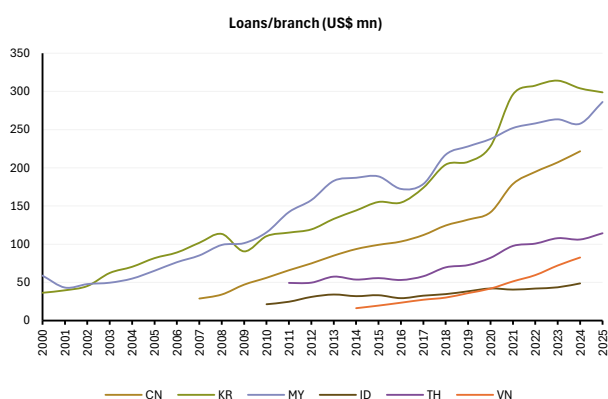
Source: Central banks, company data

Figure 43: Loans per branch (US\$m) – JP / TW / SG



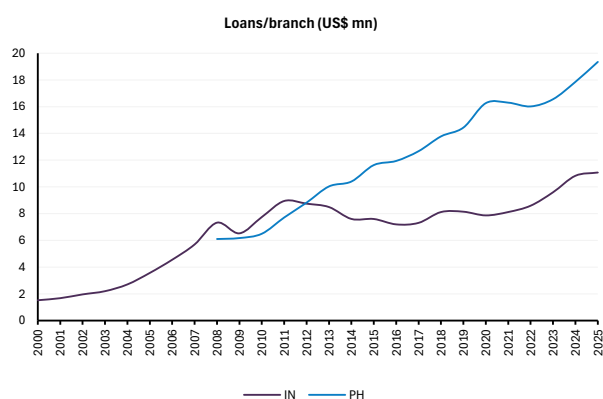
Source: Central banks, company data. Note: Data for Singapore is based on total for OCBC and UOB.

Figure 44: Loans per branch (US\$m) – CN / KR / MY / ID / TH / VN



Source: Central banks, company data. Note: Data for China is based on total for ICBC, CCB, ABC, BOC, and BOCOM. Data for Vietnam is based on total for TCB, MBB, VPB, and ACB.

Figure 45: Loans per branch (US\$m) – IN / PH



Source: Central banks, company data

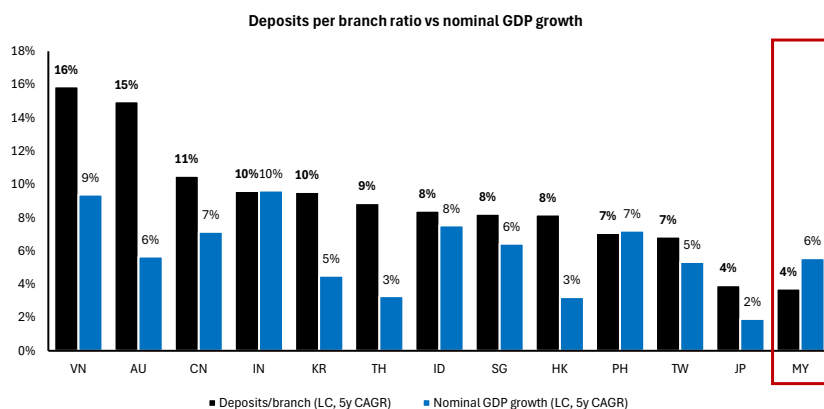
Deposits per branch ratio

Deposits per branch, measured on a LC basis, have increased across all Asian banking markets over the past five years, with Japan again the key exception due to JPY depreciation. The strongest growth was recorded in Vietnam (+16% CAGR), Australia (+15%), China (+11%), and India (+10%).

As with the loans-per-branch analysis, while deposits per branch have risen across the region, **not all markets delivered improvements relative to nominal GDP growth**. This distinction matters, as deposit growth is also closely linked to nominal economic growth; therefore, meaningful operating efficiency gains should result in deposits per branch growing faster than nominal GDP.

Our analysis indicates that all Asian banking markets achieved deposits-per-branch growth that outpaced nominal GDP growth over the past five years, except Malaysia. This suggests the Malaysian banks were less successful in improving branch level productivity and extracting operating efficiency gains on this metric.

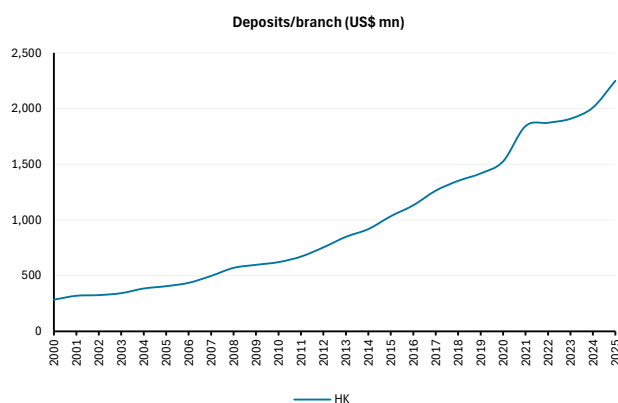
Figure 46: Growth in deposits per branch relative to nominal GDP growth



Source: Central banks, company data

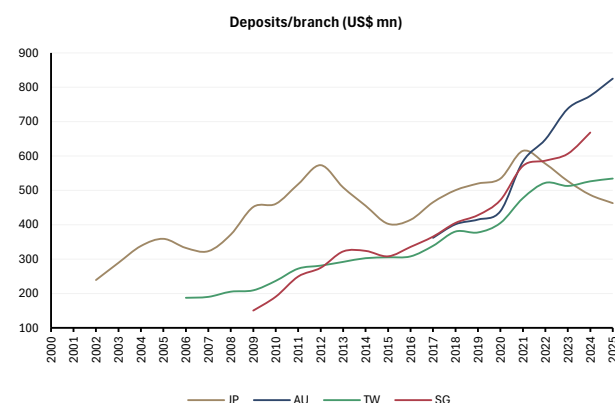
Below charts shows the absolute deposits per branch trend for the various Asian markets in US\$ terms.

Figure 47: Deposits per branch (US\$m) – HK



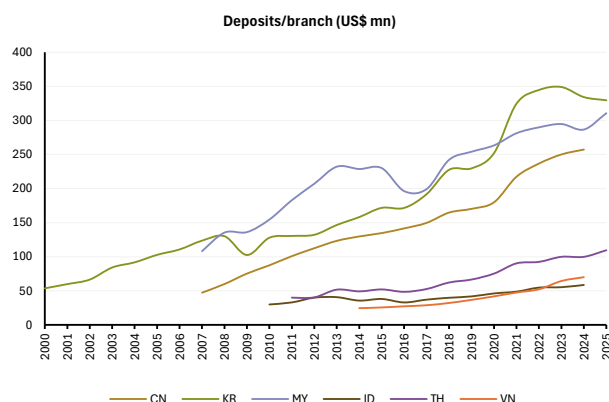
Source: Central banks, company data

Figure 48: Deposits per branch (US\$m) – JP / AU / TW / SG



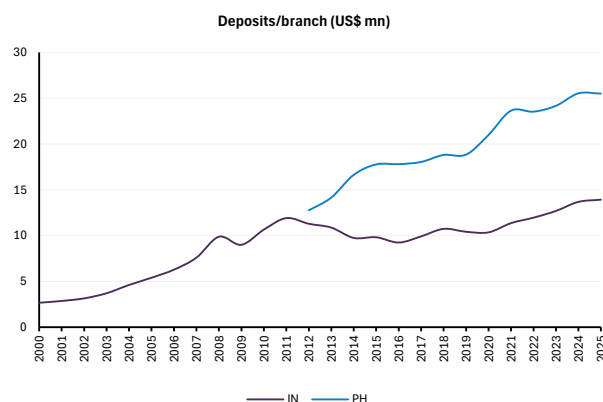
Source: Central banks, company data. Note: Data for Singapore is based on total for OCBC and UOB.

Figure 49: Deposits per branch (US\$m) – CN / KR / MY / ID / TH / VN



Source: Central banks, company data. Note: Data for China is based on total for ICBC, CCB, ABC, BOC, BOCOM, and PSBC. Data for Vietnam is based on total for TCB, MBB, VPB, and ACB.

Figure 50: Deposits per branch (US\$m) – IN / PH



Source: Central banks, company data

Will AI drive better costs efficiencies for the Asian banks?

Having discussed the key cost drivers – and the extent to which they have only partially offset structural revenue pressures for Asian banks – the question then becomes whether AI can emerge as the next structural driver of operating efficiency.

AI is increasingly viewed as a strategic necessity, with banks that fail to evolve toward AI-native operating models at risk of falling behind on both cost efficiency and competitive capability. Against this backdrop, adoption is accelerating across the region, with many Asian banks moving beyond pilot programmes toward scaled, enterprise-wide deployment of GenAI and agentic AI across core operations.

These efforts are primarily focused on three broad areas so far: 1) enhancing customer experience through personalization; 2) driving productivity gains in internal workflows like software engineering and document processing; and 3) fortifying defensive moats through real-time fraud detection and risk analytics. However, the financial impact of these use cases is uneven across categories.

From a P&L perspective, the financial impact of these initiatives is asymmetric across these three areas:

Investments in customer personalization are best viewed as defensive rather than revenue accretive, as such capabilities have increasingly become table stakes across the industry. While improved engagement and cross-sell are often cited benefits, the more tangible outcome is the prevention of customer attrition and share loss, in our view. As personalization becomes ubiquitous, its absence is more likely to result in revenue leakage than its presence driving incremental topline growth.

Productivity-oriented use cases, particularly in software development, operations, and document processing, offer clearer scope for cost efficiencies in the medium term, although quantification remains challenging at this stage. We believe gains are likely to accrue gradually through headcount cost avoidance, faster turnaround times, and reduced error rates rather than explicit cost out. As such, near-term margin uplift may be limited, with benefits manifesting more as a structural hedge against operating cost inflation over the medium term.

Lastly, AI investments in fraud detection and risk analytics have a more direct bearing on earnings stability rather than earnings expansion. While improved data ingestion and real-time analytics could modestly enhance credit underwriting outcomes, we believe the primary benefit lies in mitigating downside risks – lower fraud losses, earlier identification of asset quality deterioration, and reduced tail-risk events. In this sense, these capabilities help contain potential future costs that could otherwise materialise in the absence of such investments, supporting more resilient returns through the cycle.

As such, we believe the benefits of AI adoption are currently more operational than financial, with limited translation into reported cost ratios.

Banks are spending more on AI than they are saving, at present

Despite these early operational breakthroughs – and we would expect meaningful costs savings in the medium term given that operations can represent ~60-70% of a bank's costs base (based on the McKinsey report [here](#)) – there is currently a noticeable financialization gap where AI has not yet translated into material costs improvements in many instances. This is evident in banks commentaries (particularly HSBC) summarised in the tables below.

The key reason behind this is that while localized efficiency gains are significant – such as reducing lending decision times or automating customer messaging – these are not yet headcount-cut stories in most cases (see our discussion on headcount in the section above). They are primarily capacity-release stories (same headcount, higher throughput)

AI deployment is shifting from isolated use cases to enterprise-wide adoption

Three channels of impact: personalisation (defensive), productivity (efficiency), risk (resilience)

AI helps retain revenues more than it drives incremental growth

Efficiency = gradual, not immediate: Benefits accrue via headcount avoidance and throughput gains

Risk containment: AI improves earnings stability rather than boosting profitability.

Operational gains are real, but financial benefits remain lagged. True cost savings require scaled, end-to-end deployment.

and cost-avoidance (not hiring), so far, which shows up gradually in cost/income overtime rather than immediate opex cuts.

Furthermore, the resulting cost savings today are likely being largely reinvested in additional capital expenditure required for larger data infrastructure, technological upgrades, cybersecurity, and compliance.

Consequently, many banks report that production use cases have not yet delivered significant financial impacts, as they prioritize reinvesting these gains into further technological modernization to remain competitive against peers and digital-native disruptors.

Medium term: From implementation to costs savings

Looking ahead, we think the potential for AI to drive material cost savings for banks in the medium term remains strong given the large amount of data and human capital. Furthermore, based on the findings from McKinsey (report [here](#)), operations can represent ~60–70% of a bank's cost base, making operational AI the biggest cost lever for banks if scaled end-to-end.

For example, ICBC in China has said that it has reached a milestone where the workload handled by AI is equivalent to more than 40,000 full-time employees, illustrating the substantial scale of operational displacement possible.

Furthermore, as banks move past the initial high capex phase of building data infrastructure, these "digital workers" can scale with very low marginal cost, eventually allowing banks to scale their revenues without a linear increase in headcount or physical infrastructure.

From qualitative statements to hard numbers

As investor expectations mature, banks will need to adapt by doing more than talking about AI. The bar is rising from signalling intent to proving impact, with greater emphasis on measurable P&L outcomes and strong governance. Institutions that fail to translate investment into demonstrable financial or risk-mitigating benefits risk losing credibility, as AI narratives without execution risk being increasingly discounted by the market.

Put differently, AI is transitioning from a narrative enhancer to a delivery obligation, with execution risk becoming as relevant as technological capability. **Over time, differences in AI readiness and execution could become a key driver of cost and valuation divergence across Asian banks** (please refer to our companion APAC Focus note [here](#)).

From narrative to differentiation: AI readiness could drive future valuation dispersion

Figure 51: Tech/AI applications and implications on business performance – ex-ASEAN banks

Bank	Market	Tech/AI application and implications
ICBC	China	Built ICBC Zhiyong, a financial LLM system with over 100 billion parameters, helps to empower more than 20 main business fields and over 200 scenarios. AI wealth assistant helps to provide personalized services to its 630 million personal mobile banking users. The large-scale application of tech/AI at ICBC has resulted in a workload reduction equivalent to 40,000 persons annually.
BOC	China	Developed the BOCAI large model platform, deployed a series of large models such as DeepSeek and Qwen3, and built more than 400 intelligent assistants to empower key business functions including credit, marketing, operations, office work, customer service and technology. Leveraged enterprise-level robotic process automation (RPA) to reduce the workload of primary-level institutions, deploying RPA in over 3,600 scenarios and executing an average of nearly 300,000 tasks per month.
HSBC	Hong Kong	Have over 100 GenAI solutions in use and a strong pipeline of use cases in development. Around the globe, around 85% of HSBC staff have access to the large language model-based productivity tool, HSBC Productivity Suite, which helps them to analyse and translate documents, summarise information and generate insights. HSBC is increasing AI partnerships to accelerate adoption of cutting-edge technologies. HSBC notes that production use cases have not yet delivered "significant" financial impact in terms of bottom-line numbers, they are viewed as critical for long-term operational efficiency.
StanChart	Hong Kong	StanChart has launched SC GPT, a GenAI tool, across 41 markets worldwide. Designed to enhance operational efficiency and client engagement, SC GPT marks a significant step in the Bank's strategy to integrate AI into its business. Set to empower over 70,000 employees worldwide, SC GPT is expected to improve the Bank's operations and productivity; tailor sales and marketing efforts to increase revenue; advance software engineering with more automation; and scale how risks are measured, managed, recorded and reported. In July 2025, Standard Chartered signed a strategic partnership with Alibaba Group to deploy Alibaba Cloud AI technology across its client service, sales intelligence, risk management, and compliance functions.
Hana FG	Korea	Launched an AI-based loan product which uses AI to determine customer loan limits. The new product calculates optimal loan limits for applying customers' finances, based on its analysis of the Bank's big data of loan transaction records and patterns.
Woori FG	Korea	Use of AI to streamline work processes. For example, the AI-based market prediction system supports optimization of asset management, whilst the fraudulent transaction monitoring system also leverages AI technology. Aims to expand the use of AI to customer services, and promoting various projects to establish potential banking services in line with this goal. This includes upgrading their AI-based counseling bot and chatbot services, as well as implementing an AI Banker by developing language models tailored to the financial sector.
MUFG	Japan	Implemented AI in 116 use cases and aim to exceed 250 by FY26. MUFG is advancing the development of AI agents (e.g. co-developing an AI credit expert with Sakana AI). For employees, AI is supporting tasks such as summarizing and translating materials, creating sentences and programs, aggregating and analyzing numbers, and generating ideas. Estimates that cumulative benefits from AI use cases over its current three-year medium-term business plan will reach approximately JPY30 billion.
SMFG	Japan	Beyond improving productivity by replacing human work, SMFG aims to develop a model where AI acts as a partner, analyzing data, anticipating customer needs, and making proposals using avatars in the core of banking business such as lending and asset management.
HDFCB	India	Using cloud-based AI/ML tools, the bank decodes customer behavior to curate "hyper-personalized" content and sharpens campaign precision to improve return on investment. Actively engaging in "agentic AI" projects through its Tech Innovators platform and collaborates with the RBI Innovation Hub on AI-based early warning systems and inclusive credit tools.
Axis Bank	India	The Bank is harnessing emerging technologies such as AI/ML, automation, and data analytics to enhance customer experience, optimize operational efficiency, and improve staff productivity by building AI assists. By process Re-Engineering and digitisation, the Bank has been able to reduce the turnaround time on savings and current account opening by about 90%; and on various other processes by 50%-80%. In addition to improving efficiency, there was significant impact on customer satisfaction and risk containment.
CBA	Australia	Consistently invested in technology, data and analytics to deliver more personalised experiences for our customers and make things easier for CBA employees. CBA has achieved tangible efficiency gains, including a 65.00% improvement in recovery times from critical incidents and a 39.00% decrease in core operations complaints.
NAB	Australia	Advanced technology modernisation agenda by streamlining systems, simplifying processes, and embedding smarter, faster, and safer digital solutions. NAB's investment in AI and technology has yielded significant financial protections, preventing or recovering over \$385 million in customer scam and fraud losses in 2025.

Source: Company data

Figure 52: Tech/AI applications and implications on business performance – ASEAN banks

Bank	Market	Tech/AI application and implications
DBS	Singapore	DBS' data analytics and artificial intelligence (AI)/ machine learning (ML) initiatives delivered over SGD 750 million of economic value in 2024, more than double that in the previous year, and aiming for a total economic impact of SGD 1 billion from AI and data initiatives in 2025. The bank has implemented over 1,500 AI models across 370 distinct use cases. DBS uses AI to automate routine "white-collar" tasks, such as writing credit memos, performing KYC (Know Your Customer) checks, and screening transactions. Approximately two-thirds of staff use Gen AI for tasks like summarization, research, and translation, while data scientists use AI coding assistants to reduce coding time by up to 20%. Agentic AI is being tested to generate credit evaluation memos, and Gen AI tools summarize customer calls for quality assurance. The use of AI is expected to lead to revenue, productivity and cost-income ratio improvements.
OCBC	Singapore	AI serves as a strategic lever to maximise value in three key areas: increasing revenue, reducing risk, and enhancing productivity. About six million decisions in the Bank are made by AI on a daily basis. These tools have boosted productivity by as much as 50 per cent, freeing up employees' time and enabling them to focus on higher-value tasks. OCBC aims for 75% of all customer service requests to be AI-assisted, reducing response times and enhancing the customer experience. As workflows and processes become increasingly optimised, OCBC anticipates a 20% increase in staff productivity. Achieved close to S\$80 million in cost avoidance in 2025 through technology modernization and AI integration.
UOB	Singapore	Using GenAI to automate manual processes and increase the workflow efficiency. The end goal is to apply these AI solutions across the Bank to enhance business processes, productivity, manage risk and better personalise customer experience at scale and at speed. AI is being scaled to improve detection accuracy in areas such as Anti-Money Laundering (AML), fraud detection, and scam prevention.
Maybank	Malaysia	Piloted agentic AI in its internal auditing processes to automate selected workflows and enhance insight generation.
CIMB	Malaysia	Leverages conversational AI and chatbots to handle routine inquiries and improve response times across digital channels. AI is applied to credit decisioning, fraud detection, and general operational processes to improve consistency and service quality.
BDO	Philippines	Transitioning away from manual, low-end compliance and bookkeeping work by implementing AI solutions designed to enhance productivity, mitigate risk, and focus on higher-value advisory services.
BPI	Philippines	BPI utilizes AI and machine learning to analyze customer behavior, allowing for hyper-personalized product recommendations and "digital nudges" within its mobile apps. BPI is exploring AI-enhanced credit scoring models to better assess the creditworthiness of the "unbanked" or "underbanked" segments, using non-traditional data points to expand its lending reach.
BCA	Indonesia	Integrated the use of Augmented Reality (AR), Machine Learning (ML), and Generative Artificial Intelligence (Gen-AI) to automate internal processes and support strategic initiatives like fraud detection and AI assistance. Utilizes Gen-AI coding assistants to accelerate IT development, which significantly reduces risks and costs. Internally, AI agents at the contact center handle repetitive tasks, such as mortgage quality control, where an automatic engine replaced the manual work previously done by five agents. ML models are deployed for new customer segmentation and to predict customer attrition (churn), allowing for more targeted investment offerings.
BMRI	Indonesia	Mandiri uses AI and Gen-AI to provide personalized product recommendations and improve the efficiency of digital marketing across its digital channels. Focuses on both front-end and back-end automation to reduce operational bottlenecks and accelerate internal processes.
KBANK	Thailand	AI is embedded into end-to-end credit processes, including automated approval decisions, fraud prevention, and collection analytics to increase recovery rates. The bank has launched AINU, an AI-driven identity verification solution (facial recognition and OCR), and InsurTech solutions for automated accident assessments to process claims more efficiently.
SCBX	Thailand	Uses AI-powered advisory tools and predictive analytics to provide personalized recommendations and account planning. Its InnovestX platform delivers real-time data and AI-driven insights covering over 1,000 stocks. Digital lenders MONIX and ABACUS leverage proprietary AI engines for behavioral scoring to capture underserved segments with superior asset quality. CardX utilizes AI-driven collections to control credit costs and manage debt recovery efficiently. AI-driven security solutions identify anomalies and potential fraud by analyzing large volumes of network and user interaction data that traditional rule-based controls might miss.
TCB	Vietnam	Integrated AI across its entire value chain, processing over 8 billion data points daily to build detailed customer profiles with approximately 12,500 attributes each. Advanced analytics and machine learning initiatives delivered VND 1.3 trillion in incremental revenue in 2025. AI is being embedded into hyper-personalized credit decisioning models to enable faster, risk-aligned lending.

Source: Company data

Figure 53: Summary of AI applications by markets

Market	Top AI Priority 1	Top AI Priority 2	What this indicates about the market's AI strategy
China	Technology/engineering enablement – foundation LLM (100B parameters)	Productivity & cost efficiency – RPA across 3,600+ processes	Chinese banks emphasise technological self-sufficiency and massive scale, building proprietary large language models and deploying AI across thousands of processes to achieve huge efficiency gains.
Hong Kong	Productivity & cost efficiency – 85% of staff using GenAI tools	Revenue growth & advisory – AI-driven sales & revenue initiatives	Global banks in Hong Kong focus on enterprise-wide GenAI adoption for internal efficiency while also pursuing AI-enabled revenue growth (e.g. tailored marketing to boost sales), indicating an emphasis on long-term innovation over immediate returns.
Korea	Credit & underwriting – AI-based lending limits product	Risk, fraud & compliance – AI fraud monitoring systems	Korean banks are selectively introducing AI in core lending (e.g. automated loan limit decisions) and improving risk controls (AI-based fraud monitoring), reflecting a targeted, cautious integration of AI into traditional banking.
Japan	Productivity & cost efficiency – AI for workforce productivity (¥30 bn benefit target)	Credit & underwriting – AI credit expert in lending	Japanese banks are pursuing broad AI deployment for efficiency gains (aiming ~¥30 bn cost benefits) and innovating with AI in lending (co-developing an AI credit expert) to embed AI as a trusted partner in core banking services.
India	Customer personalisation & experience – hyper-personalised marketing & content	Productivity & cost efficiency – process automation for scale efficiency	Indian banks leverage AI to hyper-personalise products and campaigns for their huge customer bases (boosting marketing ROI) alongside aggressive automation of high-volume processes (90% faster account opening, etc.) to support growth at scale.
Australia	Customer personalisation & experience – personalised digital services	Risk, fraud & compliance – AI fraud & scam prevention	Australian banks are improving digital customer experiences (more personalised services) while strengthening AI-powered fraud and scam prevention (intercepting ~\$385 m in scams in 2025), reflecting a balance between innovation and security.
Singapore	Productivity & cost efficiency – enterprise-wide AI efficiency gains	Risk, fraud & compliance – AI in compliance & risk (KYC/AML)	Singapore's banks lead in broad AI usage, achieving significant measurable value (SGD 750 m+ in 2024) by automating tasks at scale and tightening risk/compliance controls (e.g. KYC checks, AML, fraud detection).
Malaysia	Customer personalisation & experience – AI chatbots for customer service	Risk, fraud & compliance – fraud detection & audit automation	Malaysian banks prioritise practical improvements: deploying AI chatbots to handle routine customer queries faster and starting to automate internal control processes (e.g. AI in fraud monitoring & auditing) for greater efficiency.
Philippines	Customer personalisation & experience – hyper-personalised offers & nudges	Regulatory & financial inclusion – AI credit scoring for unbanked segments	Philippine banks are catching up by automating manual tasks to improve productivity and using AI for highly tailored product suggestions & credit scoring to reach the unbanked, highlighting both modernisation and financial inclusion.
Indonesia	Productivity & cost efficiency – AR/GenAI-driven process automation	Customer personalisation & experience – personalised product targeting	Indonesian banks are mixing frontier tech (AR, GenAI) to automate internal operations (e.g. AI agents in back-office roles) with AI-driven personalisation in offerings and marketing, aligning with the country's fast-evolving digital landscape.
Thailand	Credit & underwriting – AI-embedded lending & credit processes	Revenue growth & advisory – AI-enabled advisory & fintech growth	Thai banks are applying AI in core credit operations (from loan approval to fraud checks) while developing AI-powered advisory and digital lending platforms to drive new growth and serve underserved segments.
Vietnam	Customer personalisation & experience – data-driven hyper-personalisation	Credit & underwriting – AI-augmented credit underwriting	Vietnam's banks utilise huge data sets (8 billion+ daily data points) to enable ultra-personalised customer insights and swift, AI-assisted lending decisions, supporting strong revenue growth in loans.

Source: Company data, UBS

How do banking markets compare on opex-related metrics?

In this section, we summarise the key metrics across the Asian banking markets discussed above. The table below provides an overview of how the various revenue drivers (as a percentage of assets) have trended over the past 10 years across markets.

Figure 54: Opex drivers (as % of assets)

As % of assets (10y chg)	Staff costs	Other opex	Total opex
China	↓	↓	↓
Hong Kong	→	↓	↓
Korea	↓	↓	↓
India	↑	↑	↑
Japan	↓	↓	↓
Australia	→	↓	↓
Taiwan	↓	↑	↑
Singapore	→	↓	→
Malaysia	↓	↑	↑
Philippines	↓	→	→
Indonesia	↓	↓	↓
Thailand	↓	→	↓
Vietnam	↓	↓	↓
Avg (Asia)	↓	↓	↓

Source: Central banks, company data, UBS estimates

Net interest income/assets ratio

The table on the right shows the absolute change in net interest income/assets ratio for the various markets over the past 5y/10y/15y. Over the past 10 years, c50% of Asia banking markets have seen a decline in net interest income/assets ratio.

Markets where the net interest income/assets ratios have declined most were China, followed by Indonesia, and Malaysia. On the other hand, markets that have seen an increase in the net interest income/assets ratio were Philippines, India, and Singapore.

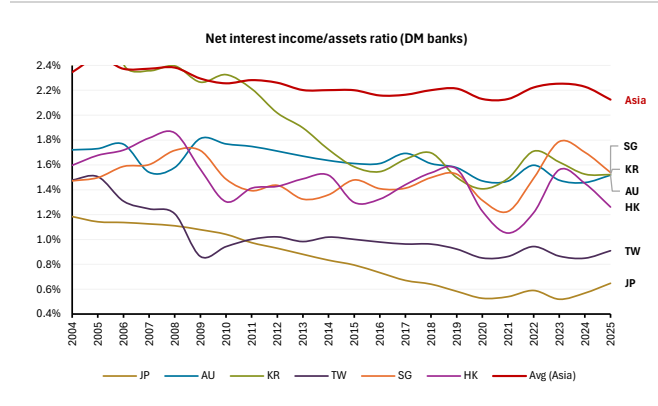
Below charts shows the net interest income/assets trend for the banking markets over the past 20 years. As of 2025, banking markets with the lowest net interest income/assets ratio were Japan (0.65%), Taiwan (0.91%), and China (1.24%). Markets with the highest net interest income/assets ratio were Indonesia (4.84%), the Philippines (3.91%), and Vietnam (3.11%).

Figure 55: Net interest income/assets (chg)

NII/assets	5y	10y	15y
Japan	0.1%	-0.1%	-0.4%
Australia	0.0%	-0.1%	-0.3%
China	-0.7%	-1.1%	-1.0%
Korea	0.1%	-0.1%	-0.8%
Taiwan	0.1%	-0.1%	0.0%
Singapore	0.2%	0.1%	0.1%
Philippines	0.4%	1.0%	0.7%
India	0.3%	0.5%	0.7%
Thailand	0.1%	0.0%	-0.2%
Malaysia	0.0%	-0.2%	-0.6%
Hong Kong	0.0%	0.0%	0.0%
Indonesia	0.0%	-0.9%	-0.6%
Vietnam	-0.8%	0.0%	3.1%
Avg (Asia)	0.0%	-0.1%	-0.1%

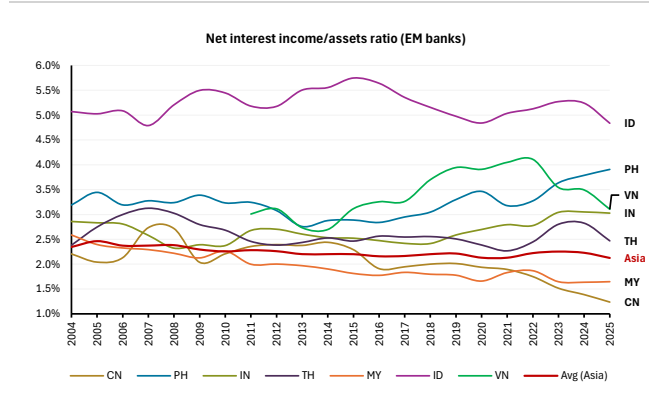
Source: Central banks, company data, UBS estimates

Figure 56: Net interest income/assets ratio (DM)



Source: Central banks, company data, UBS estimates

Figure 57: Net interest income/assets ratio (EM)



Source: Central banks, company data, UBS estimates

Fee income/assets ratio

The table on the right shows the absolute change in fee income/assets ratio for the various markets over the past 5y/10y/15y. Over the past 10 years, c50% of Asia banking markets have seen a decline in fee income/assets ratio.

Markets where the fee income/assets ratios have declined most were China, followed by Thailand, and Hong Kong. On the other hand, markets that have seen an increase in the fee income/assets ratio were Vietnam and India.

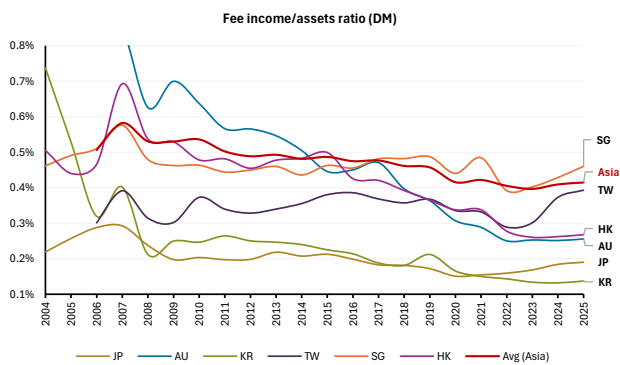
Below charts shows the fee income/assets trend for the banking markets over the past 20 years. As of 2025, banking markets with the lowest fee income/assets ratio were Korea (0.14%), Japan (0.19%), and China (0.24%). Markets with the highest fee income/assets ratio were Indonesia (1.05%), the Philippines (0.61%), and India (0.60%).

Figure 58: Fee income/assets (chg)

Fees/assets	5y	10y	15y
Japan	0.0%	0.0%	0.0%
Australia	-0.1%	-0.2%	-0.4%
China	-0.1%	-0.3%	-0.3%
Korea	0.0%	-0.1%	-0.1%
Taiwan	0.1%	0.0%	0.0%
Singapore	0.0%	0.0%	0.0%
Philippines	0.1%	0.0%	-0.1%
India	0.0%	0.1%	-0.1%
Thailand	-0.1%	-0.3%	-0.1%
Malaysia	0.0%	-0.1%	-0.2%
Hong Kong	-0.1%	-0.2%	-0.2%
Indonesia	0.0%	-0.1%	0.0%
Vietnam	0.1%	0.3%	0.4%
Avg (Asia)	0.0%	-0.1%	-0.1%

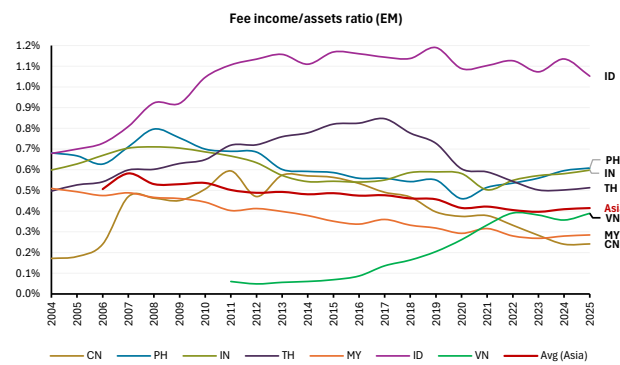
Source: Central banks, company data, UBS estimates

Figure 59: Fee income/assets ratio (DM)



Source: Central banks, company data, UBS estimates

Figure 60: Fee income/assets ratio (EM)



Source: Central banks, company data, UBS estimates

Total operating expenses/assets ratio

The table on the right shows the absolute change in total opex/assets ratio for the various markets over the past 5y/10y/15y. Over the past 10 years, c50% of Asia banking markets have seen a decline in total opex/assets ratio.

Markets where the total opex/assets ratios have declined most were China, followed by Indonesia, and Vietnam. On the other hand, markets that have seen an increase in the total opex/assets ratio were India, Malaysia, and Taiwan.

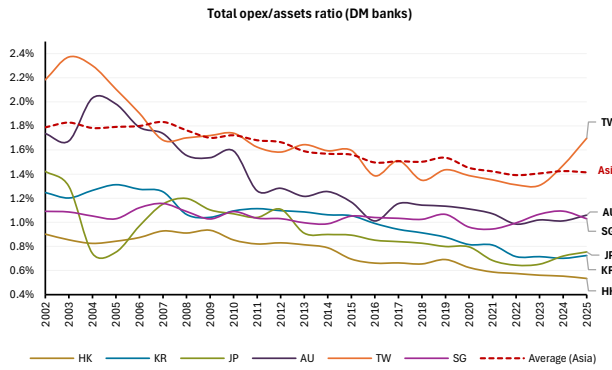
Below charts shows the total opex/assets trend for the banking markets over the past 20-25 years. As of 2025, banking markets with the lowest total opex/assets ratio were Hong Kong (0.53%), and China (0.58%), and Korea (0.72%). Markets with the highest total opex/assets ratio were the Philippines (2.48%), Indonesia (2.42%), and India (2.07%).

Figure 61: Total opex/assets (chg)

Total opex/assets	Latest (2025)	5y	10y	15y
Philippines	2.48%	0.0%	0.0%	-0.8%
Indonesia	2.42%	-0.2%	-0.6%	-0.8%
India	2.07%	0.1%	0.3%	0.3%
Malaysia	1.96%	0.2%	0.2%	0.0%
Taiwan	1.70%	0.3%	0.1%	0.0%
Thailand	1.60%	-0.1%	-0.2%	-0.2%
Avg (Asia)	1.40%	0.0%	-0.2%	-0.3%
Vietnam	1.34%	-0.5%	-0.5%	-
Australia	1.06%	-0.1%	-0.1%	-0.5%
Singapore	1.03%	0.1%	0.0%	-0.1%
Japan	0.75%	0.0%	-0.1%	-0.3%
Korea	0.72%	-0.1%	-0.3%	-0.4%
China	0.58%	-0.2%	-0.6%	-0.6%
Hong Kong	0.53%	-0.1%	-0.2%	-0.3%

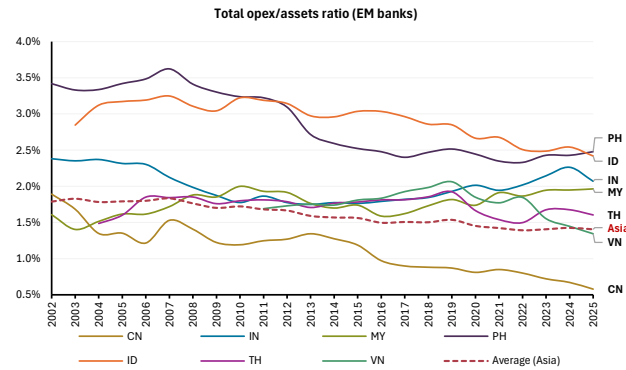
Source: Central banks, company data, UBS estimates

Figure 62: Total operating expenses/assets ratio (DM)



Source: Central banks, company data, UBS estimates

Figure 63: Total operating expenses/assets ratio (EM)



Source: Central banks, company data, UBS estimates

Staff costs/assets ratio

The table on the right shows the absolute change in staff costs/assets ratio for the various markets over the past 5y/10y/15y. Over the past 10 years, most banking markets have seen a decline in staff costs/assets ratio.

Markets where the staff costs/assets ratios have declined most were China, followed by Indonesia, and Vietnam. On the other hand, India was the only markets that saw an increase in the staff costs/assets ratio.

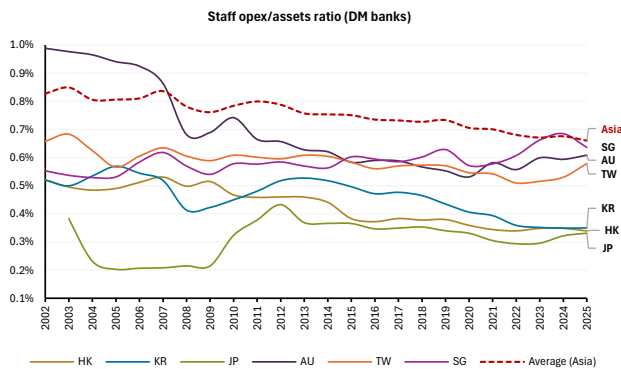
Below charts shows the staff costs/assets trend for the banking markets over the past 20-25 years. As of 2025, banking markets with the lowest staff costs/assets ratio were China (0.29%), Japan (0.33%), and Hong Kong (0.34%). Markets with the highest staff costs/assets ratio were Indonesia (1.36%), India (1.03%) and the Philippines (0.90%).

Figure 64: Staff costs/assets (chg)

Staff costs/assets	Latest (2025)	5y	10y	15y
Indonesia	1.36%	-0.2%	-0.3%	-0.3%
India	1.03%	0.0%	0.1%	0.0%
Philippines	0.90%	0.0%	-0.1%	-0.3%
Malaysia	0.74%	0.0%	-0.1%	-0.2%
Vietnam	0.73%	-0.3%	-0.2%	-0.1%
Thailand	0.70%	-0.1%	-0.1%	-0.1%
Avg (Asia)	0.66%	0.0%	-0.1%	-0.1%
Singapore	0.64%	0.1%	0.0%	0.1%
Australia	0.61%	0.1%	0.0%	-0.1%
Taiwan	0.58%	0.0%	0.0%	0.0%
Korea	0.35%	-0.1%	-0.1%	-0.1%
Hong Kong	0.34%	0.0%	0.0%	-0.1%
Japan	0.33%	0.0%	0.0%	0.0%
China	0.29%	-0.1%	-0.2%	-0.3%

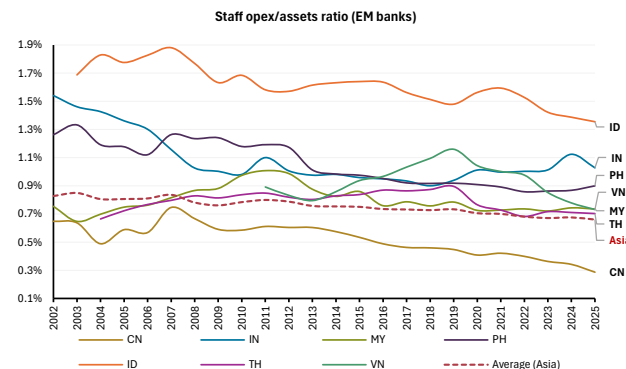
Source: Central banks, Company data, UBS estimates

Figure 65: Staff costs/assets ratio (DM)



Source: Central banks, company data, UBS estimates

Figure 66: Staff costs/assets ratio (EM)



Source: Central banks, company data, UBS estimates

Other opex/assets ratio

The table on the right shows the absolute change in other opex/assets ratio for the various markets over the past 5y/10y/15y. Over the past 10 years, c50% of banking markets have seen a decline in other opex/assets ratio negatively impacted by lower policy rates.

Markets where the other opex/assets ratios have declined most were China, Indonesia, and Vietnam. On the other hand, markets that have seen an increase in the other opex/assets ratio were Malaysia, India, and Taiwan.

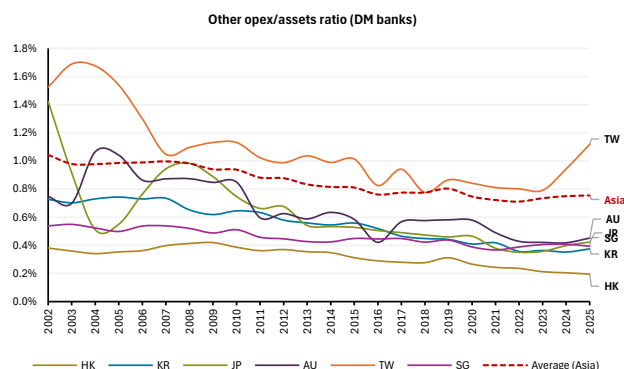
Figure 67: Other opex/assets (chg)

Other opex/assets	Latest (2025)	5y	10y	15y
Philippines	1.58%	0.0%	0.0%	-0.5%
Malaysia	1.23%	0.2%	0.3%	0.2%
Taiwan	1.12%	0.3%	0.1%	0.0%
Indonesia	1.06%	0.0%	-0.3%	-0.5%
India	1.04%	0.0%	0.2%	0.2%
Thailand	0.90%	0.0%	0.0%	-0.1%
Avg (Asia)	0.74%	0.0%	-0.1%	-0.2%
Vietnam	0.61%	-0.2%	-0.3%	-0.3%
Australia	0.45%	-0.1%	-0.1%	-0.4%
Japan	0.42%	0.0%	-0.1%	-0.3%
Singapore	0.39%	0.0%	-0.1%	-0.1%
Korea	0.37%	0.0%	-0.2%	-0.3%
China	0.29%	-0.1%	-0.4%	-0.3%
Hong Kong	0.20%	-0.1%	-0.1%	-0.2%

Below charts shows the other opex/assets trend for the banking markets over the past 20-25 years. As of 2025, banking markets with the lowest other opex/assets were Hong Kong (0.20%), China (0.29%), and Korea (0.37%). Markets with the highest other opex/assets were the Philippines (1.58%), Malaysia (1.23%), and Taiwan (1.12%).

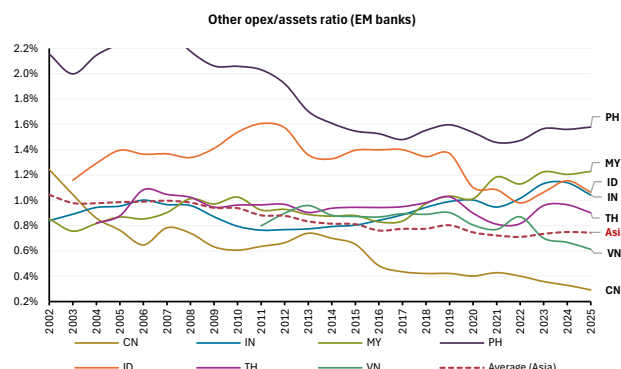
Source: Central banks, company data, UBS estimates

Figure 68: Other opex/assets ratio (DM)



Source: Central banks, company data, UBS estimates

Figure 69: Other opex/assets ratio (EM)



Source: Central banks, company data, UBS estimates

Staff costs/total opex ratio

The table on the right shows the absolute change in staff costs/total opex ratio for the various markets over the past 5y/10y/15y. Over the past 10 years, most banking markets have seen an increase in staff costs/total opex ratio.

Markets where the staff costs/total opex ratios have declined the most were India, Thailand, and Philippines. On the other hand, markets that have seen the largest increase in the staff costs/total opex ratio were Hong Kong, Australia, and China.

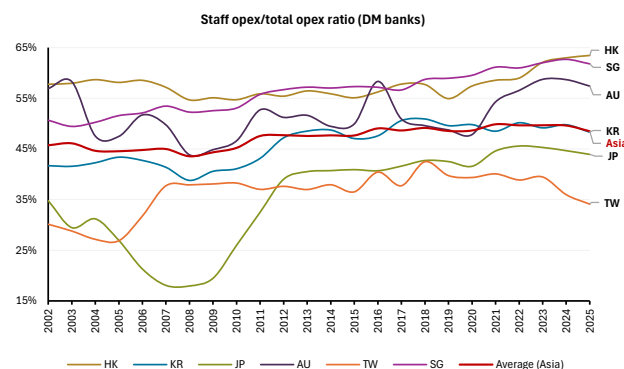
Below charts shows the staff costs/total opex trend for the banking markets over the past 20-25 years. As of 2025, banking markets with the lowest staff costs/total opex ratio were Taiwan (34%), Philippines (36%), and Thailand (44%). Markets with the highest staff costs/total opex ratio were Hong Kong (63%), Singapore (62%), and Australia (57%).

Figure 70: Staff costs/total opex (chg)

Staff opex/total opex	Latest (2025)	5y	10y	15y
Hong Kong	63%	6%	8%	9%
Singapore	62%	2%	4%	9%
Australia	57%	10%	7%	11%
Vietnam	54%	-2%	3%	
Malaysia	52%	1%	2%	3%
India	50%	-1%	-5%	-6%
China	50%	-1%	5%	0%
Avg (Asia)	49%	1%	2%	4%
Korea	48%	-2%	1%	7%
Indonesia	45%	0%	-1%	-1%
Japan	44%	2%	3%	18%
Thailand	44%	-2%	-3%	-3%
Philippines	36%	-1%	-2%	0%
Taiwan	34%	-5%	-2%	-4%

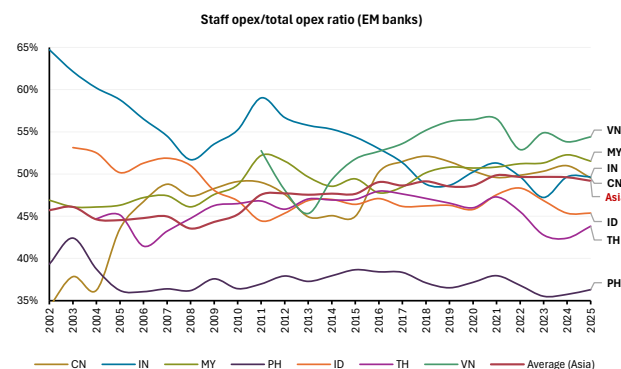
Source: Central banks, company data, UBS estimates

Figure 71: Staff costs/total opex ratio (DM)



Source: Central banks, company data, UBS estimates

Figure 72: Staff costs/total opex ratio (EM)



Source: Central banks, company data, UBS estimates

Total opex/revenue ratio

The table on the right shows the absolute change in total opex/revenue ratio for the various markets over the past 5y/10y/15y. Over the past 10 years, most banking markets have seen a decline in total opex/revenue ratio.

Markets where the total opex/revenue ratios have declined most were Vietnam, Korea, and Philippines. On the other hand, markets that have seen the largest increase in the total opex/revenue ratio were Japan, Australia, and India.

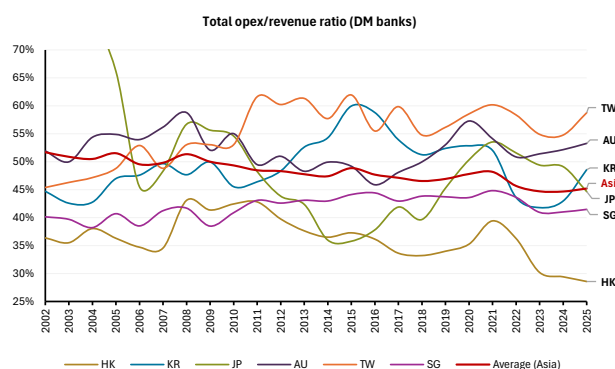
Below charts shows the total opex/revenue trend for the banking markets over the past 20-25 years. As of 2025, banking markets with the lowest total opex/revenue ratio were Hong Kong (29%), Vietnam (30%), and China (38%). Markets with the highest total opex/revenue ratio were Japan (63%), Taiwan (59%), and Australia (53%).

Figure 73: Total opex/revenue (chg)

Total opex/revenue	Latest (2025)	5y	10y	15y
Japan	63%	-4%	4%	4%
Taiwan	59%	1%	-3%	6%
Australia	53%	-4%	4%	-2%
Philippines	53%	0%	-10%	-8%
Indonesia	50%	-14%	-8%	-6%
Korea	49%	-4%	-11%	3%
India	48%	0%	1%	3%
Malaysia	47%	3%	0%	2%
Avg (Asia)	46%	-3%	-4%	-3%
Thailand	43%	-4%	-6%	-11%
Singapore	41%	-2%	-3%	1%
China	38%	6%	0%	-2%
Vietnam	30%	-4%	-14%	
Hong Kong	29%	-7%	-9%	-14%

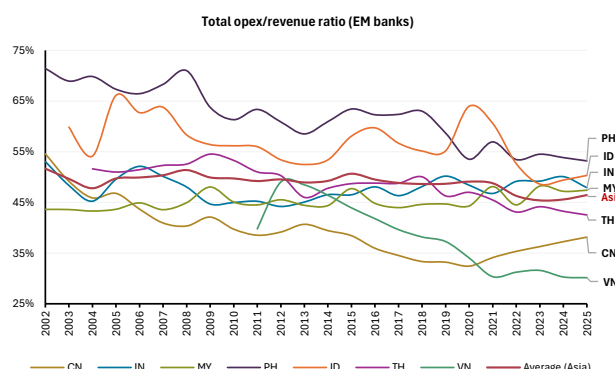
Source: Central banks, company data, UBS estimates

Figure 74: Total opex/revenue ratio (DM)



Source: Central banks, company data, UBS estimates

Figure 75: Total opex/revenue ratio (EM)



Source: Central banks, company data, UBS estimates

Staff costs/revenue ratio

The table on the right shows the absolute change in staff costs/revenue ratio for the various markets over the past 5y/10y/15y. Over the past 10 years, most banking markets have seen a decline in staff costs/revenue ratio.

Markets where the staff costs/revenue ratios have declined most were Philippines, Vietnam, and Korea. On the other hand, markets that have seen the largest increase in the staff costs/revenue ratio were Australia, China, and Japan.

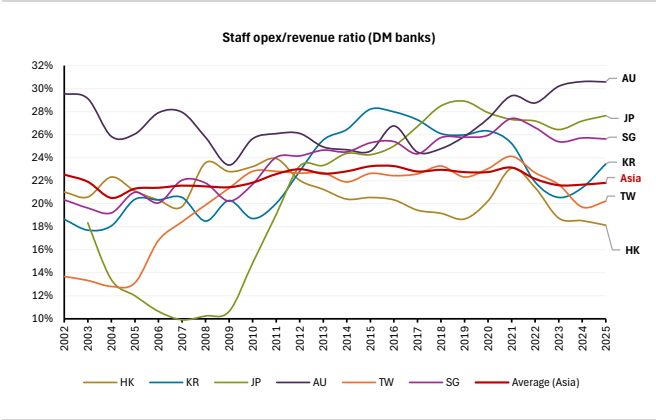
Below charts shows the staff costs/revenue trend for the banking markets over the past 20-25 years. As of 2025, banking markets with the lowest staff costs/revenue ratio were Vietnam (16%), Hong Kong (18%), and Thailand (19%). Markets with the highest staff costs/revenue ratio were Australia (31%), Japan (28%), and Singapore (26%).

Figure 76: Staff costs/revenue (chg)

Staff opex/revenue	Latest (2025)	5y	10y	15y
Australia	31%	3%	6%	5%
Japan	28%	0%	3%	13%
Singapore	26%	0%	0%	4%
India	24%	-1%	-2%	-1%
Korea	23%	-3%	-5%	5%
Malaysia	22%	0%	-1%	0%
Avg (Asia)	22%	-1%	-1%	0%
Taiwan	20%	-3%	-2%	-3%
Philippines	19%	-1%	-5%	-3%
China	19%	3%	2%	-1%
Indonesia	19%	-3%	-2%	-3%
Thailand	19%	-3%	-4%	-6%
Hong Kong	18%	-2%	-2%	-5%
Vietnam	16%	-3%	-6%	

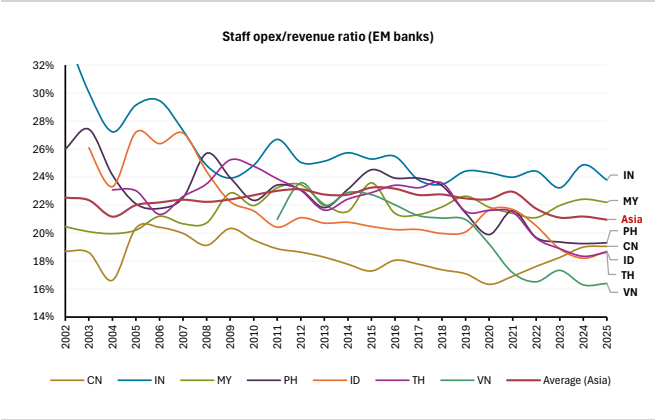
Source: Central banks, company data, UBS estimates

Figure 77: Staff costs/revenue ratio (DM)



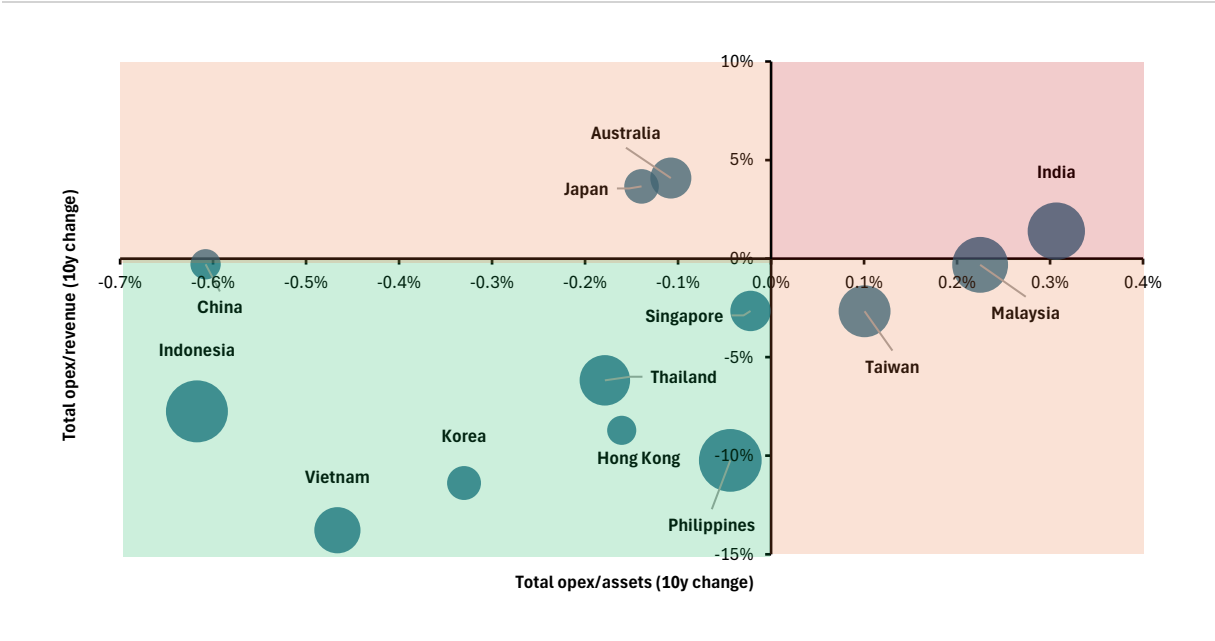
Source: Central banks, company data, UBS estimates

Figure 78: Staff costs/revenue ratio (EM)



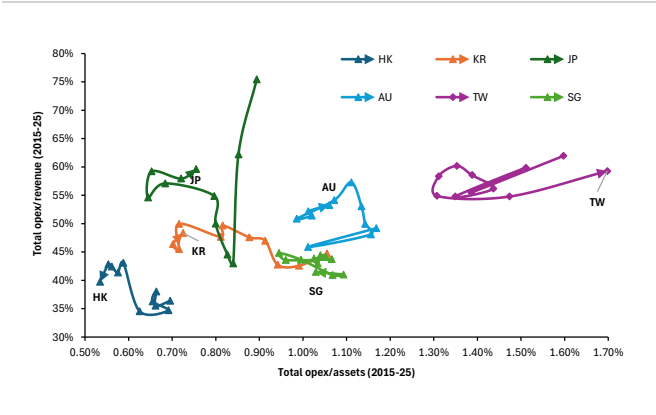
Source: Central banks, company data, UBS estimates

Figure 79: Opex changes in the past 10 years across Asian banks (bubble size indicates the current level)



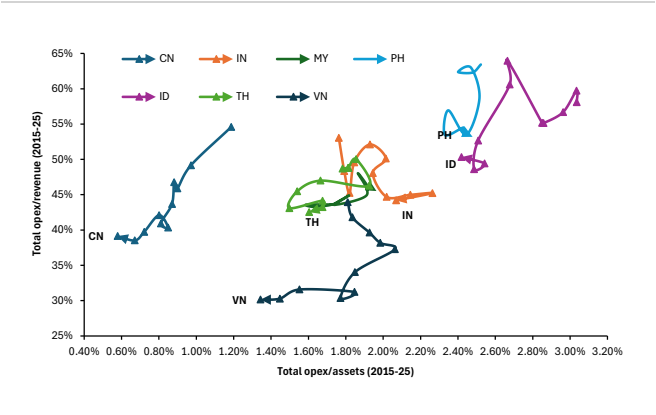
Source: UBS estimates. Note: Bubble sizes are scaled to banks' respective total opex/assets ratios as of 2025.

Figure 80: How has opex/revenue and opex/assets ratio changed over the past 10y (DM)?



Source: Central banks, company data, UBS estimates

Figure 81: How has opex/revenue and opex/assets ratio changed over the past 10y (EM)?



Source: Central banks, company data, UBS estimates

Valuation Method and Risk Statement

The main risk for Asian banks is inflation in countries with current account deficits and low FX reserves as higher interest rates would hurt margins for banks with high LDRs and hurt credit quality. Also, asymmetric increases in deposit rates and fewer rises in lending rates could hurt current high loan margins. Banks with equity/asset ratios below 5% and ROAs below 1% are vulnerable to economic and regulatory risk, especially if still growing loans by more than 15% per annum, as they will encounter capital shortages.

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Neutral	FSR is between -6% and 6% of the MRA.	40%	21%
Sell	FSR is > 6% below the MRA.	6%	21%

Source: UBS. Rating allocations are as of 31 March 2026.

1:Percentage of companies under coverage globally within the 12-month rating category.

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